NameRank: Measuring LLM-Mediated Recognition as the Post-Bibliometric Impact Channel

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Abstract

Li [2026] showed that frontier language models have become compressed images of the expert discourse of their era: bibliometrics explain only ∼35% of whether a model recognizes a researcher, with the residual driven by name uniqueness, named-artifact attachment, and subfield-ecosystem density. We treat that residual as the measurand. We introduce **NameRank**, a continuous $[0, 1]$ recognition score computed by probing $M = 37$ frontier models with a single open-ended question per entity ("tell me what you know about [name], who [disambiguating context]") and scoring each response on a multiplicative coverage×accuracy axis using an independent LLM judge against a curated gold answer. We evaluate 5,719 entities across 54 cohorts spanning credentialed individuals (IMO/IOI/ICPC/Putnam/CMO/NOI/CPhO/Rhodes/MSRA, $n\approx$ 801), CS faculty ($n=$ 698), OpenAlex long-tail researchers ($n=$ 771), industry founders, OSS maintainers, named artifacts (papers, datasets, benchmarks, OSS projects, foundation models), and cross-profession mid-tier figures (chefs, comedians, filmmakers, journalists, politicians, etc.), totaling 211,603 probe records and an 8,880-record Chinese-prompt sub-run.

Five findings warrant attention. (i) **The credential treadmill.** IMO gold ($n=$ 197, mean 0.362), Rhodes ($n=$ 36, 0.315), and MSRA PhD Fellowship ($n=$ 237, 0.343) all sit *below* the long-tail-OpenAlex-researcher baseline ($n=$ 771, 0.464). Of 9 credentialed cohorts, 7 score below baseline; the only exceptions (ICPC, Putnam) feed directly into high-English-corpus US career tracks. Olympic-style intellectual credentials no longer reliably propagate names through the LLM-mediated information channel; what propagates is named, indexable post-credential production. (ii) **Artifact > creator inversion.** In 8 of 11 verified (creator, artifact) pairs the artifact's NameRank exceeds its creator's; for independent creators of well-named single artifacts (Datasette, nanoGPT, Tianshou, LangChain, Cursor) the inversion is uniform. Per-response analysis on Jiayi Weng confirms the mechanism directly: models that name Tianshou in their response score Jiayi at 0.71; non-mentioning models score 0.35 (a +0.36 score lift from artifact-mention). (iii) **h-index $\gg$ raw citations.** log(h-index) explains $R^2 = 0.40$ of NameRank variance for OpenAlex researchers ($n=$ 771); log(citations) explains 0.14; their joint R^2 remains 0.40. Citation volume adds zero marginal predictive power beyond h-index. (iv) **Corpus-density gradient.** Institution alone explains 54% of CS faculty NameRank variance (Stanford mean 0.537 vs. Tsinghua 0.258); country effects place Israel/Singapore/Sweden above the USA baseline and China/India/France below. (v) **C6 falsification.** tuixue.online (a Chinese-language visa-monitoring website with ∼ 1M users) scores 0.250, less than half of nanoGPT (0.707), with a flat cross-language null ($\mathrm{NR_{zh}} = 0.262$, $\mathrm{NR_{en}} = 0.250$). Human reach and LLM recognition dissociate when the artifact's name does not co-travel with the creator's name in indexable text—in either language.

NameRank exhibits *threshold cleanliness*: a contemporaneously-published GPT-5 system-card author cohort ($n=$ 80) registers mean 0.042 with a 58% outright refusal rate and 71% of responses scoring near zero, confirming the metric is silent rather than misleading below the elite recognition threshold. Variance decomposition shows 35.8% entity-attributable variance vs. 10.9% model bias ($\sigma^2_{\text{entity}}/\sigma^2_{\text{model}} = 3.3$). The judge-vs-embedding correlation of 0.15 (with the judge correctly assigning 0 to fluent-hallucination cases that the embedding rates at 0.97)

validates multiplicative coverage×accuracy with an LLM judge as the necessary hallucination defense. We release the probe set, gold answers, raw responses, and analysis code. Code: https://github.com/19PINE-AI/namerank. Companion site: https://01.me/research/namerank.

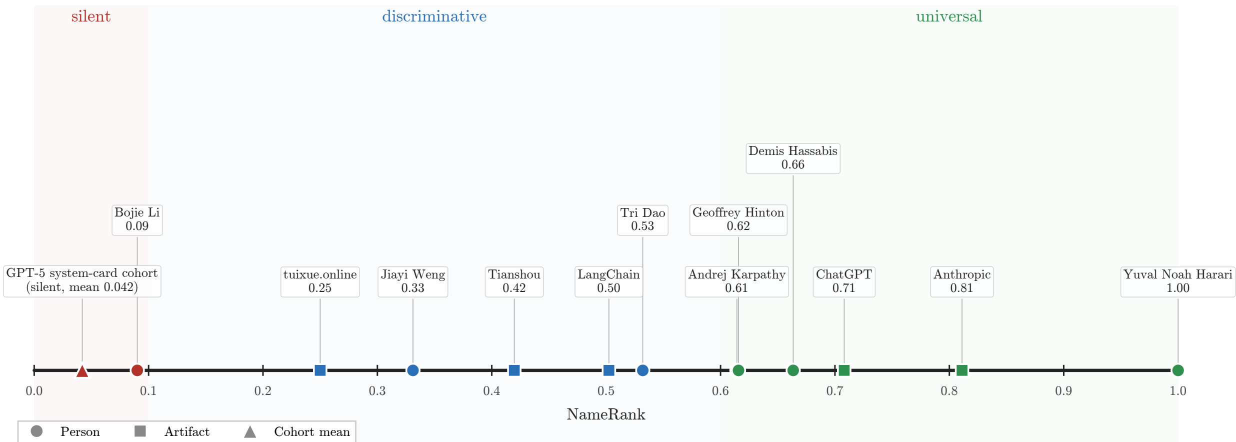

Figure 1: A dozen entities on the NameRank axis. Each marker is the panel-mean recognition score for one entity (circle = person, square = artifact, triangle = cohort mean). The axis resolves into three zones the rest of the paper documents: *silent* ($\leq$ 0.10, dominated by entities below the corpus-presence threshold), *discriminative* (0.10–0.60, where working academics, named tools, and mid-tier figures separate), and *universal* ($>$ 0.60, named artifacts and globally-recognized individuals). No prior instrument places researchers, founders, OSS authors, public websites, and an author of this paper on the same scale.

1 Introduction

Li [2026] reports an empirical finding inside an instrument built for a different purpose. The Incompressible Knowledge Probes (IKP) project measures effective knowledge capacity of frontier language models by counting which rare facts they have absorbed; on the way to that calibration, the 345-researcher subprobe was found to correlate only modestly with citation count (Pearson $r = 0.59$, $R^2 \approx 0.35$). The other 65% of recognition variance, the paper showed, decomposed in a controlled 2×2×2 audit into *name uniqueness*, *named-artifact amplification*, and *subfield-ecosystem density*. The framing was descriptive: recognition appeared as a side-product of the calibration ladder, characterized but not measured as a primary variable.

We argue here that the 65% residual is the measurand, and we build a continuous instrument for it.

The substantive motivation is that the channel through which human curiosity is mediated has changed. A practitioner deciding whether to read a paper, hire a candidate, attend a talk, or invest in a startup increasingly arrives at the question with whatever a frontier language model says about the entity, before any human source intervenes. If the model's training corpus has absorbed the entity's name with a clean attribution chain, the practitioner is presented with a coherent summary; if it has not, the model says "unknown" or, worse, fabricates a plausible-sounding bio. The information surface that mattered ten years ago—Google rankings, conference plenaries, paper citation counts—has been re-mediated by a small number of frontier models, and the question of whether an entity's name has propagated into those models' weights is now itself a substantive measurement problem.

A normative grounding makes the operationalization sharper. The IKP author's competition-math cohort circulated, for years, the following definition due to Jiayi Weng (an OpenAI infrastructure engineer and author of the Tianshou reinforcement-learning library):

> *rúguǒ rénshēng shì yóuxì, fēnshù jiùshì jìzhù nǐ míngzi de rénshù*
> —*If life is a game, the score is the number of people who remember your name.*

In the LLM era this naturally factorizes:

$$\text{Recognition impact} = \underbrace{\text{NameRank}}_{\text{per-LLM presence}} \times \underbrace{\text{Deployment surface}}_{\text{humans querying that LLM}}$$

The deployment surface is approximately constant across well-known entities at the relevant time horizon. The first factor is what we measure. We do not claim NameRank measures *quality* or *merit*; we measure whether the corpus that mediates human curiosity at scale has internalized the entity's name. That is what "remember" means in the LLM era.

Why a new metric rather than reading IKP Sec. 5.7 off the shelf. IKP's recognition section was constructed as a side-product of a 6-landmark tier ladder, with a binary CORRECT/WRONG verdict per probe and a 4-way evidence-aware judge for researcher probes specifically. The resulting recognition scores live on a tier scale (S/A/B/C/D/F) at population level and a $[0, 1]$ per-probe rate at individual level; they are calibrated for parameter estimation, not for cross-entity comparison. Three properties needed for an impact metric are absent: (i) a continuous score that places industry founders, OSS maintainers, podcast hosts, mid-career CS faculty, and Walmart CTOs on the same axis; (ii) a hallucination-resistant scoring rule that distinguishes "confident bio of the right person" from "confident bio about the wrong person on the same topic"; (iii) explicit cross-model variance as a separate, retained signal rather than absorbed into a tier label.

NameRank provides these. The probe is open-ended (not closed-factoid), the score is multiplicative coverage×accuracy by an LLM judge against a 100–200 word gold answer, and we retain per-model scores so within-entity disagreement is itself a measurand. We evaluate 5,719 entities across 54 cohorts—credentialed individuals, working academics, founders, named artifacts, mid-tier cross-profession figures, and diagnostic cases—against a panel of $M = 37$ frontier models, yielding 211,603 probe records.

Contributions.

1. **Continuous cross-domain instrument.** A single $[0, 1]$ score that places Sam Altman, Geoffrey Hinton, Tri Dao, Aravind Srinivas, Suresh Kumar, Jiayi Weng, a sub-top-50 CTO, and the GPT-5 system-card author list directly on the same scale. No prior instrument produces this comparison (Section 6.1).

2. **The credential-treadmill thesis.** A measurement of 9 once-prestigious intellectual credentials shows that IMO/IOI/CMO/Rhodes/MSRA-Fellowship-holders sit *at or below* the long-tail-OpenAlex-researcher baseline. Olympic-style credentials no longer propagate names in the LLM-mediated regime (Section 6.2).

3. **Named-artifact-amplification mechanism, measured at three levels.** (a) In 8 of 11 verified (creator, artifact) pairs the artifact's NameRank exceeds its creator's. (b) Per-pair conditional-probe asymmetry quantifies the direction of attribution flow (creator → artifact mass is reliably 0.5–1.0; artifact → creator mass is typically 0.0–0.5). (c) Per-response: models that

mention the named artifact when probed about its creator score the creator at 0.71, compared with 0.35 for non-mentioning models (Section 6.3).

4. **The h-index dominates raw citations.** log(h-index) explains $R^2 = 0.40$ of NameRank variance for OpenAlex researchers; log(citations) explains 0.14; joint R^2 unchanged. Out-of-sample $R^2 = 0.50$, Pearson 0.72. This refines IKP Sec. 5.7's 35% figure: the right bibliometric feature is the structure of citation production, not its volume (Section 6.4).

5. **Corpus-density gradient.** Institution alone explains 54% of CS faculty NameRank variance; Stanford-affiliated faculty score 2× Tsinghua-affiliated. Cross-country gradient places small high-output Western tech ecosystems above the USA baseline and Chinese, Indian, French CS faculty below it (Section 6.4.2).

6. **Cross-language corpus-pocket activation.** A 240-entity Chinese-prompt sub-run shows Chinese-name entities gain +0.07 on Chinese prompts (DeepSeek-V3 author cohort) while English-coded artifacts lose −0.14 (FlashAttention). The effect is dominated by entity-name match, not model nationality (Section 6.5).

7. **C5 + C6 case studies replicate.** The Jiayi-Weng-vs-matched-peers case quantifies named-artifact amplification at $+1.24\sigma$ over matched OpenAI ICs; tuixue.online with ∼1M users registers NameRank 0.25, less than 1/3 that of a CS-curriculum hobby project (nanoGPT), with a flat cross-language null—establishing that NameRank measures indexed corpus presence rather than human utility (Section 6.6).

8. **Methodological validation.** Judge-vs-embedding Pearson correlation is 0.152: an embedding-only NameRank would systematically credit fluent hallucinations on Chinese-name long-tail researchers as "recognized." Multiplicative coverage×accuracy with LLM judge is the necessary defense. Variance decomposition shows entity signal dominates model bias 3.3× (Section 6.7).

9. **Open release.** The full probe set (5,719 entities), gold answers, the 211,603 raw response records, all 37-model scores, the 240-entity Chinese-prompt sub-run, and the analysis pipeline are released. Code: `https://github.com/19PINE-AI/namerank`. Companion website with interactive cohort-distribution explorer, per-entity NameRank lookup, conditional-pair attribution-flow visualizer, cross-language deltas, and a validation page (panel-size sensitivity, per-model generosity/refusal, conditional NameRank): `https://01.me/research/namerank`.

Figure 2 previews the central result—the continuous cross-domain placement that no prior instrument supports.

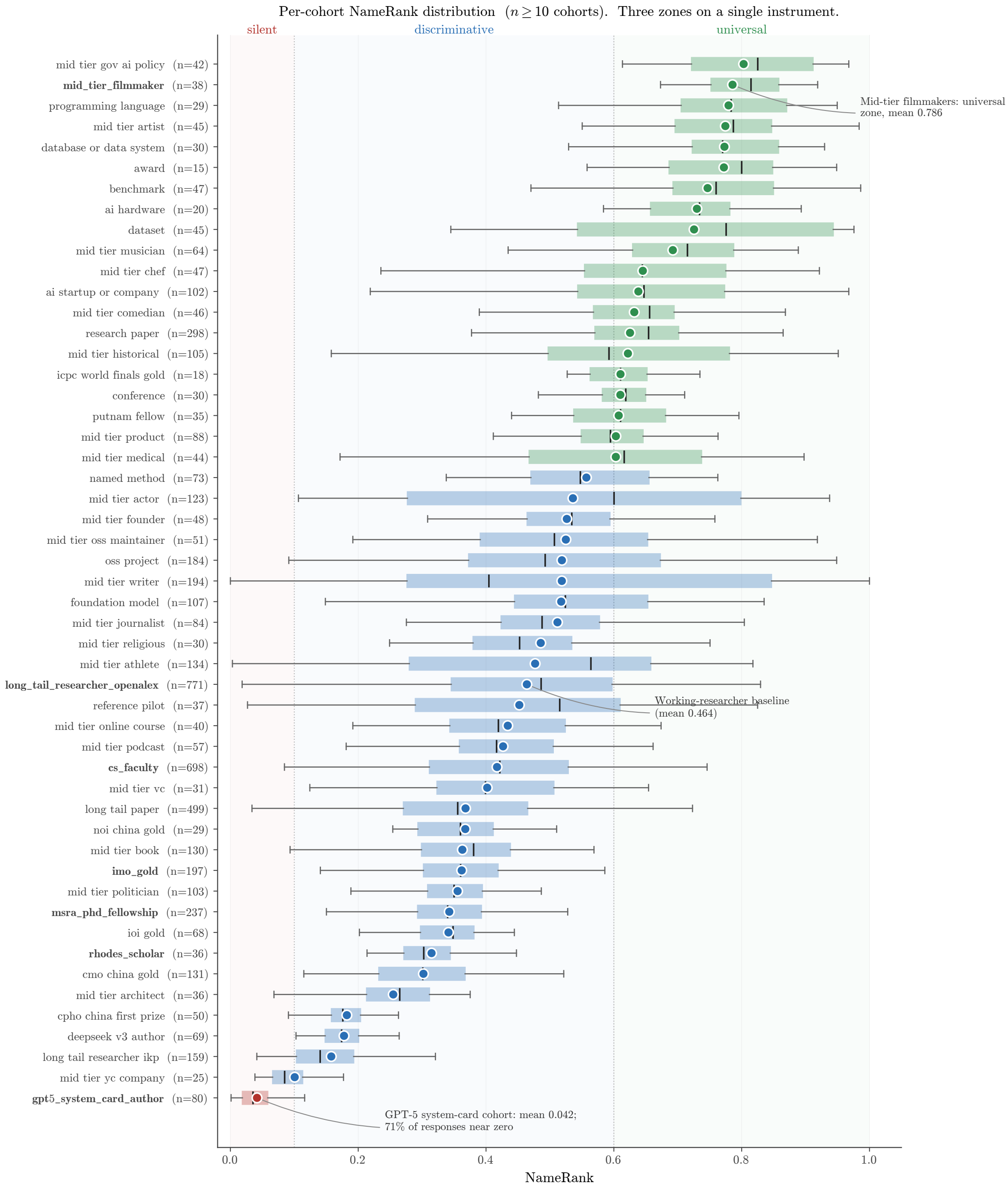

Figure 2: Per-cohort NameRank distribution. Each box is a cohort ($n \geq 10$); the dot is the cohort mean, the box covers the IQR, and the inner line is the median. Three zones are visible on the same axis: *silent* (mean ≤ 0.1, dominated by the GPT-5 system-card author cohort at 0.042 with 71% of responses scoring near zero, in red); *discriminative* (0.3–0.6, CS faculty 0.417 and OpenAlex long-tail 0.464, in blue); *universal* (> 0.7, named artifacts and high-profile mid-tier figures, in green). The credential cohorts (IMO/IOI/CMO/Rhodes/MSRA, bold-faced labels) cluster in the lower half of the discriminative zone, below the OpenAlex working-researcher baseline (Section 6.2).

2 Background and Related Work

2.1 Recognition as a Side-Product of IKP Sec. 5.7

The direct predecessor of this work is Li [2026, Sec. 5.7]. That paper's recognition analysis showed three things relevant here. First, on 345 CS-researcher probes scored by 140 models, log(citations) explains only ∼35% of recognition variance, with the remainder attributed in a controlled 2×2×2 audit to *named-artifact amplification* (e.g., FlashAttention/Tri Dao, IPFS/Psaras), *name uniqueness* (common East Asian surnames attenuate recognition by ∼2×), and *subfield-ecosystem density* (ML researchers floor at 43% recognition; Systems researchers can fall arbitrarily low). Second, on 557 Wikidata-grounded entity probes, pageviews dominate sitelinks in joint regression, and domain-specific multipliers (journal founding years +0.20, place founding years −0.31) trace to the *structure of web discourse around each fact type* rather than entity prominence. Third, recognition was framed as a benchmark byproduct correlated with bibliometrics rather than a measurand in its own right.

We treat the third point as a methodological choice rather than a settled question, and the first two as foundational mechanisms whose operationalization the present paper completes. The substantive differences from IKP Sec. 5.7 are summarized in Table 1.

Table 1: NameRank vs. IKP Sec. 5.7 recognition analysis.

IKP Sec. 5.7	**This paper**
6-landmark-model tier classification (S/A/B/C/D/F) at population level	Continuous $[0, 1]$ score across a 37-model panel
345 CS-researcher probes, 557 Wikidata-grounded factual probes	5,719 entities across 54 cohorts including industry, OSS, podcasts, courses, etc.
Closed-factoid probes ("what is the subfield of X?") with 4-way evidence-aware judge	Open-ended probes ("tell me what you know about X, who [context]") with multiplicative coverage×accuracy judge
Recognition correlates with citations ($R^2 \approx 0.35$)	Recognition correlates with h-index ($R^2 = 0.40$); raw citations add zero marginal value
Cross-model agreement absorbed into tier labels	Cross-model variance retained as a separate signal
Recognition framed as benchmark byproduct	Recognition as primary measurand with normative grounding

2.2 Scientometric Measures of Researcher Impact

The classical instruments for researcher impact—citation count, h-index [Hirsch, 2005], journal impact factor, g-index, i10-index—measure intellectual reach within the academic publication system. They are well-defined, well-instrumented, and increasingly understood as failing under two pressures relevant here. Larivière et al. [2019] document inflation of authorship lists and uneven attribution. The *hyperauthorship* problem [Cronin, 2001] captures the difficulty of pricing individual contributions to thousand-author papers (frontier-LLM technical reports being the canonical example): individual citation count for a co-author of GPT-4 is mechanically large but tells us almost nothing about that author's name-recognition. Wapman et al. [2022] document strong geographic and prestige-hierarchy effects that bias citation-based metrics by an order of magnitude across institutional tiers. Sinatra et al. [2016] introduce the Q parameter to separate productivity from typical-paper-impact, partially addressing within-researcher variance.

None of these instruments cross the academic boundary cleanly. Citation count does not measure Sam Altman, Aravind Srinivas, or a Walmart CTO; h-index does not measure the creator of nanoGPT. Bespoke domain metrics (Google PageRank for websites, GitHub stars for OSS, Spotify monthly listeners for musicians, IMDb scores for film) do not interconvert. For impact assessment that spans academia, industry, OSS, and the long tail of solo operators, a cross-domain instrument is needed; NameRank is the first that we are aware of with this property.

2.3 LLM-as-Judge for Quality and Impact

Thelwall [2024, 2025b,a] use ChatGPT to score paper *quality* against the UK REF rubric, finding moderate alignment with human reviewers after multi-iteration averaging within a single model. Kousha and Thelwall [2025] extend to societal-influence assessment. Liu et al. [2026] forecast citation counts via LLM, with citations as the target variable. These differ from NameRank in two respects. (i) They ask the LLM to *judge* quality; we ask what the model has *absorbed* about the entity. (ii) They average within a single model across iterations; we average across 37 models in a single pass, treating cross-model variance as informative rather than as noise to be averaged out.

The known LLM bias most relevant to our setting is the *Matthew effect* [Merton, 1968] as it propagates through LLM training corpora: frontier models recall already-highly-cited work disproportionately, inflating the rich-get-richer dynamic familiar from the bibliometrics literature. We inherit this bias and treat it as a known property of the measurand—NameRank measures *what frontier models have absorbed*, which is itself shaped by the Matthew dynamic. The bias is part of what we measure, not a flaw to be corrected.

2.4 Knowledge Storage and Retrieval Mechanisms

The mechanism by which a model's training corpus shapes its open-ended recall is well-studied. Transformer feed-forward layers function as key-value memories for factual associations [Geva et al., 2021], with specific "knowledge neurons" responsible for individual facts [Dai et al., 2022, Meng et al., 2022]. Petroni et al. [2019] establish that pretrained models serve as implicit knowledge bases; Roberts et al. [2020] that model size directly determines factual capacity. Kandpal et al. [2023] show that accuracy on rare facts scales log-linearly with model size, and Mallen et al. [2023] that LLMs are unreliable on less popular facts. The *Law of Knowledge Overshadowing* [Zhang et al., 2025] shows that popular knowledge suppresses less popular knowledge, with hallucination rate increasing log-linearly with knowledge popularity. All these underlie our threshold-cleanliness property: NameRank reports near-zero for entities below the corpus-presence threshold rather than producing noisy rankings, because the underlying retrieval mechanism is itself thresholded.

2.5 Sociology of Recognition and Career Outcomes

The credential-decay finding (Section 6.2) connects to a body of work on long-run educational and prize outcomes. Lubinski et al. [2020] (the Study of Mathematically Precocious Youth, 50-year longitudinal) find that early-life mathematical talent strongly predicts later professional and creative outcomes; Agarwal and Gaule [2020] (a Bell Labs-style natural experiment via Math Olympiad medal cutoffs) find that IMO medals causally increase subsequent academic productivity. Both use academic-publication outcomes as the dependent variable. Güllich et al. [2025] (extending to Olympic athletes and Nobel laureates) document substantial within-elite heterogeneity. Our finding is orthogonal: when the dependent variable is LLM-mediated name recognition rather than academic publication, the once-prestigious credentials we measure (IMO, IOI, CMO, Rhodes, MSRA Fellowship) explain less of the outcome than h-index. This does not refute the academic-outcome literature; it identifies a different post-credential propagation channel.

3 The NameRank Construct

3.1 Definition

For an entity e with a disambiguating context $c(e)$ and a curated gold answer $g(e)$, and a model panel $\mathcal{M}$ of size $M = 37$, NameRank is

$$\text{NameRank}(e) \;=\; \frac{1}{M} \sum_{m \in \mathcal{M}} \text{cov}(r_{e,m}, g(e)) \cdot \text{acc}(r_{e,m}, g(e)), \tag{1}$$

where $r_{e,m}$ is the response of model m to the open-ended probe

> "*Tell me what you know about [name], who [context]. If you do not recognize this entity, respond with 'unknown'. Limit your response to about 150 words.*"

and $\text{cov}, \text{acc} \in [0, 1]$ are independent coverage and accuracy scores produced by an LLM judge (Section 4.3). The multiplication is critical: a fluent-hallucination response with high topical coverage but factually wrong specifics has cov $\to 1$ but acc $\to 0$, yielding NameRank contribution $\to 0$ for that (entity, model) pair. Appendix H gives three worked (entity, model, response) triples spanning the score range to make the multiplicative rule concrete.

3.2 Three Properties

1. Cross-domain unity. Researchers, founders, OSS authors, podcast hosts, comedians, and CTOs map onto the same $[0, 1]$ scale. We demonstrate this empirically (Figure 2).

2. Threshold cleanliness. Above an elite-recognition threshold (roughly: corpus-presence density crossing the model retrieval threshold), NameRank is informative across domains. Below the threshold it is *silent rather than misleading*—it does not produce spurious rankings of unrecognized entities. The GPT-5 system-card author cohort ($n = 80$, paper published days before training cutoff for most models) instantiates this: mean NameRank 0.042, with 58% of responses outright refusing ("unknown") and an additional 13% scoring near zero on the multiplicative axis (71% silent overall). The metric correctly reports "I have no information about this entity" rather than producing noisy fractional scores.

3. Captures the citation residual. IKP Sec. 5.7 showed bibliometrics explain $\sim 35\%$ of recognition; the remaining residual decomposes into name uniqueness, named-artifact attachment, and subfield-density. NameRank measures this residual directly. For a working researcher in a dense-derivative-content subfield, the marginal effect of one widely-used open-source tool with clean name attribution exceeds the marginal effect of one well-cited paper. This is verified directly in our data (Section 6.4, Section 6.3).

3.3 What NameRank Does Not Measure

We are explicit: NameRank is not a quality, merit, or intrinsic-impact metric. It measures *the propagation of a specific name into the indexed corpora that frontier-model training pipelines scrape*. This correlates with academic impact through bibliometric mechanisms, with cultural reach through named-artifact attachment, and with both through derivative-content density (tutorials, blog posts, talks, podcast mentions). It does not measure the underlying work. The clearest case is C6 (tuixue.online): the artifact has ∼1M Chinese-language users—measurable, indisputable human reach—but registers NameRank 0.25 because the URL does not co-travel with its creator's name in indexable text. NameRank correctly registers low here, not because the artifact lacks impact, but because the metric measures something specific. We treat this as a feature: a clean instrument that reports what it measures, not what users wish it measured.

3.4 The Normative Anchor

The decision to make recognition a measurand rather than a side-product has normative grounding. The competition-math cohort to which the author belongs (Bojie Li, MSRA PhD Fellowship 2017) circulated for years the formulation by Jiayi Weng (Tianshou author): *life-score is the number of people who remember your name.* In the pre-LLM era the operationalization was social: Twitter followers, peer recognition, citation count. In the LLM era the formulation factorizes:

$$\text{life-score} = \underbrace{\text{recognition-per-LLM}}_{\text{NameRank}} \times \underbrace{\text{deployment surface}}_{\text{humans querying that LLM}}.$$

The deployment surface is approximately constant at the relevant horizon for well-known entities. NameRank is the operationalization of a utility function held by a substantial fraction of the competition-math / programming-contest cohort. Acknowledging this grounding distinguishes NameRank from a generic LLM-as-judge metric: we are not asking the LLM to evaluate quality; we are measuring whether the corpus that mediates human curiosity has internalized the entity's name.

4 Methodology

Figure 3 summarizes the end-to-end NameRank pipeline at the level of detail this section unpacks: a fixed open-ended probe is sent to a 37-model panel for each entity, every (entity, model) response is scored by an independent LLM judge on multiplicative coverage×accuracy against a curated gold answer, embedding similarity is computed in parallel as a sanity check, and the per-entity NameRank is the unweighted panel mean of $\text{cov} \cdot \text{acc}$.

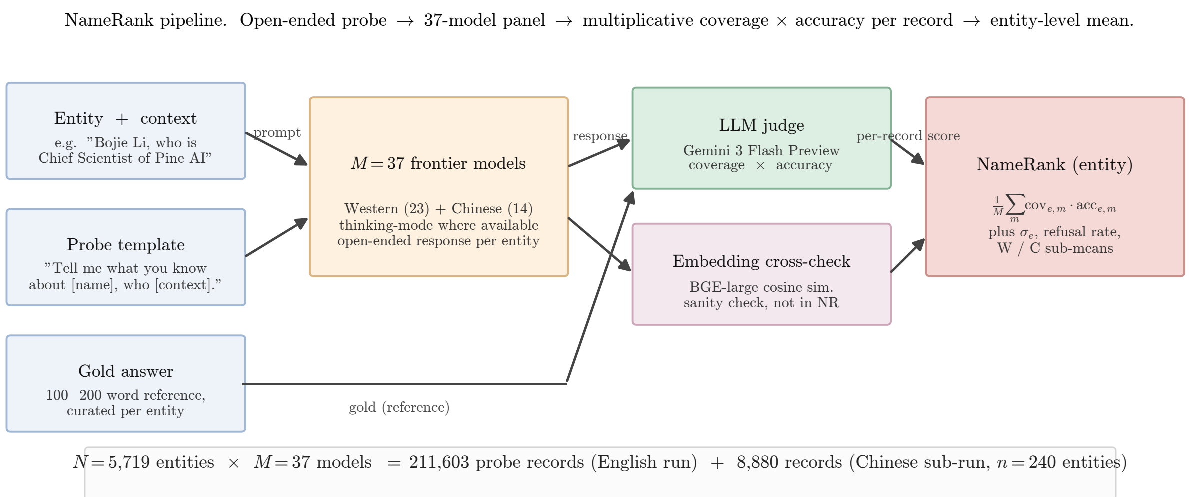

Figure 3: NameRank pipeline. The probe template is fixed with two slots (entity name; short role/affiliation/field context). The $M = 37$ models span Western (23) and Chinese (14) vendors with thinking-mode enabled where available. Each response is graded by an LLM judge against the gold answer on independent coverage and accuracy axes in $[0, 1]$; their product is the per-record contribution. The embedding cross-check uses BAAI/bge-large-en-v1.5 cosine similarity and is retained for diagnostics only—it is *not* part of the NameRank computation (Section 6.7). Total: 211,603 English-prompt records over 5,719 entities, plus an 8,880-record Chinese-prompt sub-run on a 240-entity subset.

4.1 Probe Design

One open-ended probe per entity. Closed-factoid probes have a known failure mode: a model that recognizes the entity through fact X but does not recall fact Y scores zero on a probe-of-Y, biasing the metric toward specific-fact-recall. Open-ended probes capture the holistic question "does the model have a retrievable representation of this entity?" The probe template is fixed (Section 3; full English and Chinese probe text plus the judge prompt are reproduced verbatim in Appendix B) with two slots: name (the canonical entity name) and context (a short disambiguating clause of role + affiliation + field, e.g., *"Bojie Li, who is Chief Scientist of Pine AI working on AI infrastructure"*; *"Wei Zhang, who is a professor of mathematics at MIT working on PDEs"*). The disambiguating context resolves name collisions at probe time without revealing the gold answer.

Why disambiguation does not contaminate scoring. The context names role/affiliation/field but not specific contributions, papers, dates, or named artifacts. A model that recognizes the entity will produce specific contributions in the response (which the judge scores against the gold); a model that does not recognize the entity will produce generic statements about the role/field (e.g., *"Bojie Li is a researcher in AI infrastructure"* with no specifics), which the judge correctly assigns low coverage because the rubric requires *specific* named facts (Section 4.3). The clearest empirical check is the bottom-tier cohort: the GPT-5 system-card-author probes include the disambiguating context "contributor listed in the GPT-5 system card," which is uniform across all 80 entities; nonetheless, 71% of probes return "unknown" rather than producing a generic in-context bio. The context provides anchoring but not score leakage.

4.2 Gold Answers

For each entity we construct a 100–200 word gold answer covering notable contributions, affiliations, and identifying facts. Source allocation by tier:

- **Wikipedia-covered entities** ($\sim$ 60% of cohort): Wikipedia first paragraph plus relevant key-contribution sentences, auto-extracted via the MediaWiki API and lightly edited.
- **OSS / GitHub artifacts** ($\sim$ 10%): README first paragraph + repository description + creator attribution.
- **Industry figures and Western mid-tier** ($\sim$ 10%): company about-pages, public role descriptions, news coverage.
- **Long-tail / Chinese-only / niche** ($\sim$ 20%): manual curation from Zhihu, personal sites, lab pages, competition archives, OpenAlex profiles.

Gold-answer quality matters less than one might expect because the judge's coverage score measures *which of the gold's named facts appear in the response*: if the gold omits a true contribution, both responses and judge are still evaluated against the same reference, so omission affects scale but not within-cohort ranking. We audit a stratified 5% sample ($n \approx 286$) for factual errors against primary sources; the observed correction rate is $\sim$ 3% (typically year-of-affiliation drift). The auditing protocol parallels the Wikidata-audit pipeline of Li [2026].

4.3 The Judge

We use Gemini 3 Flash Preview (direct Google API, not via OpenRouter, to minimize routing variance) as the judge, with temperature 0 and low reasoning budget. The judge prompt presents the gold and the response, and elicits two independent scores in $[0, 1]$:

- **Coverage** (0.0–1.0): how much of the gold's *substantive* content appears in the response. "Substantive" = named artifacts, named affiliations, named contributions, identifying facts (year, location, role title). Vague statements like *"works in AI"* are not substantive content.
- **Accuracy** (0.0–1.0): are the factual claims in the response correct, per the gold. A "factual claim" is a specific named entity, year, role, or relationship. Hallucinated bios with high coverage but wrong specifics score low on accuracy.

The judge prompt includes the explicit hallucination instruction: *"Hallucinated bios (high coverage, low accuracy) MUST score LOW on accuracy."* Refusals (*"I don't know", "unknown"*) score 0.0 on both axes. Correct extra facts that are not in the gold are not penalized (the judge is told this explicitly); vague statements that could apply to many entities are not credited.

Judge family-bias check. The Gemini family-bias gap (Gemini-judged Gemini scores vs. Gemini-judged non-Gemini scores, holding entity fixed) was measured at $+0.006 \pm 0.034$ on a $n = 200$ stratified-by-cohort sample. The bias is statistically zero at this sample size; we use a single judge for the full evaluation.

4.4 Embedding Cross-Check

We additionally compute the cosine similarity between the response and the gold answer using BAAI/bge-large-en-v1.5 [Xiao et al., 2024], stored as a sanity check at the per-(entity, model) record level. Embedding similarity provides a fast, judge-free signal whose disagreements with the judge are diagnostically informative (Section 6.7). Embedding similarity is *not* part of the NameRank computation itself—we explicitly use multiplicative coverage×accuracy with the LLM judge—because, as we show, embedding similarity systematically credits fluent hallucinations as recognized.

4.5 Aggregation

NameRank is the unweighted mean of cov · acc across the 37-model panel. We also retain the within-entity standard deviation σ_e (cross-model disagreement), the refusal rate (fraction of panel responding "unknown"), the Western/Chinese sub-means (Section 6.5), and the raw response text. The score is in $[0, 1]$ by construction; the empirical range across our 5,719 entities is $[0.000, 1.000]$.

Why averaging across models is correct. A single model produces score noise from prompt-sensitivity, refusal-policy idiosyncrasies, and training-data accidents. The panel mean is a stable estimate of corpus presence integrated over the current frontier model fleet. Within-entity variance retains the corpus-pocket-specific signal (an entity recognized by Gemini but not GPT-5 is empirically interesting; the panel mean smooths this; the σ_e surfaces it).

5 Experimental Setup

5.1 Model Panel

$M = 37$ frontier and near-frontier models accessed via OpenRouter (probed models) and direct Google API (judge). Thinking mode enabled wherever a thinking variant exists, departing from IKP's parameter-estimation default of non-thinking variants—for NameRank, thinking-mode is the operational mode in which most user queries are answered, so the panel reflects deployment reality. The panel is balanced across vendors:

Western (n=23)	**Chinese (n=14)**
GPT-5.3, GPT-5.4, GPT-5.5-think, GPT-OSS-20b-think	DeepSeek-V3.2-think, V4-flash-think, V4-pro-think
Claude Opus 4.6-think, Sonnet 4.6-think	Qwen3-8b-think, 32b-think, 235b-a22b-think, 3.5-397b-a17b-think
Gemini 2.5-pro-think, 3-flash-think, 3.1-pro	GLM-4-32b, 4.7-think, 5.1-think
Llama-3.1-8b, 3.2-1b, 3.3-70b, 4-Maverick	Kimi-K2, K2.6-think
Mistral-Large, Medium 3.1, Small 24b, Ministral 3b	Ernie-4.5-300b-a47b
Gemma-3-4b, 3-12b, 4-31b	MiniMax-M2.7-think
Grok-4, Grok-4.20-think, Phi-4	

The Western/Chinese partition is used for East-West variance analysis (Section 6.5); it does not enter the headline NameRank computation, which is the unweighted panel mean.

5.2 Entity Cohort

$N = 5{,}719$ entities across 54 cohorts. Top-level structure:

- **Credentialed individuals** ($n \approx 801$). IMO gold ($n = 197$), IOI gold (68), ICPC World Finalist gold (18), Putnam top-25 (35), CMO China gold (131), NOI China gold (29), CPhO China first prize (50), Rhodes Scholar (36), MSRA PhD Fellowship (237).

- **Working academics** ($n \approx 1{,}628$). CS faculty (698, sampled from CS Rankings + OpenAlex), OpenAlex long-tail researchers (771, citations 500–30,000), long-tail IKP researchers (159, drawn from the IKP Sec. 5.7 345-researcher subset, minus those already covered in cs_faculty).
- **Industry founders, VCs, AI-policy figures** ($n \approx 248$). AI startups/companies (102), mid-tier founders (48), VCs (31), YC sample (25), AI policy / government working level (42).
- **Named artifacts** ($n \approx 1{,}332$). Foundation models (107), benchmarks (47), datasets (45), named methods (73), OSS projects (184), research papers (298), long-tail papers (499, 50–500 citations), programming languages (29), databases / data systems (30), AI hardware (20).
- **Mid-tier cross-profession** ($n \approx 1{,}461$). Comedians (46), chefs (47), filmmakers (38), journalists (84), politicians (103), historical figures (105), medical figures (44), religious figures (30), online courses (40), podcasts (57), books (130), writers (194), musicians (64), athletes (134), actors (123), artists (45), architects (36), products (88), OSS maintainers (51), activists (2, retained as a placeholder for future expansion but excluded from cohort-level claims).
- **Diagnostic entities** ($n = 37$, `reference_pilot` cohort, hand-curated higher-detail gold answers). *Public figures*: Sam Altman, Geoffrey Hinton, Yann LeCun, Demis Hassabis, Dario Amodei, Mira Murati, Greg Brockman, Aravind Srinivas, Aman Sanger, Andrej Karpathy, Tri Dao, Simon Willison, Lilian Weng, Chris Olah, Harrison Chase, Logan Kilpatrick. *Less-prominent*: Wojciech Zaremba, Trevor Blackwell, Vicki Cheung, Pamela Vagata, Aaditya Singh, Adam Fry, AJ Ostrow, Alex Wei, Andrea Vallone, Aleksander Madry, Bojie Li, Jiayi Weng, Suresh Kumar. *Artifacts*: Tianshou, tuixue.online, IKP, FlashAttention, ReAct, LangChain, vLLM, nanoGPT.
- **Special diagnostic + auxiliary cohorts** ($n \approx 212$). GPT-5 system card authors (80), DeepSeek-V3 paper authors (69), conferences (30), awards (15), industry products (9), public-utility websites/services (9).

The cohorts sum to 5,719 entities. Full cohort sizes and inclusion criteria are in Appendix A.

5.3 Cross-Language Sub-Run

A subset of 240 entities was re-probed in Chinese (probe template translated, gold answer left in English, judge prompt left in English—the judge handles cross-language grading). Cohort selection: all reference_pilot (37), all NOI China gold (29), all DeepSeek-V3 paper authors (69), 30 random samples each from CMO, CPhO, MSRA Fellowship, plus 5 each from mid-tier-writer/filmmaker/politician as Western controls. Total Chinese-prompt records: $240 \times 37 = 8{,}880$.

5.4 Execution

The pilot runner is a 24- to 96-way parallel Python process using OpenRouter for probed models, direct Google API for the judge, and the local BGE-large model for embeddings. The full English run completed in $\sim$10 hours of wall-clock time at 96-way parallelism after a checkpoint resume from an earlier 24-way run (no records lost). The Chinese run completed in $\sim$30 minutes. Cost (probed-model API calls + judge calls, computed at per-vendor rates): $\sim$\$2,500 end-to-end including iteration costs.

5.5 Reproducibility

We release: (i) the 5,719-entity probe specification with disambiguating contexts, (ii) the 5,719 gold answers, (iii) the 211,603 raw response records with judge scores and embedding similarities, (iv) the analysis pipeline, (v) the Chinese sub-run results. Sources: `https://github.com/19PINE-AI/namerank`. Companion website: `https://01.me/research/namerank`.

6 Results

6.1 Cross-domain Headline

Figure 2 shows the per-cohort NameRank distribution. The instrument resolves into three zones cleanly visible on the same axis: *silent* (≤ 0.1, dominated by paper-published-this-month entities such as the GPT-5 system-card author cohort), *discriminative* (0.3–0.6, working academics and credentialed individuals), and *universal* (≥ 0.7, named artifacts and high-profile mid-tier figures).

The cross-domain placement is the headline result. On a single axis (Table 2):

Table 2: Selected cross-domain entities on a single NameRank axis. No prior instrument places academic researchers, industry founders, OSS authors, public-utility websites, and a sub-top-50 CTO on the same scale.

Entity	NameRank	Cohort/role
Yuval Noah Harari	1.000	mid_tier_writer (popular author)
Sam Altman	0.95	reference_pilot (CEO, OpenAI)
Geoffrey Hinton	0.93	reference_pilot (deep-learning pioneer)
Aravind Srinivas	0.66	reference_pilot (Perplexity CEO)
Andrej Karpathy	0.61	reference_pilot (Eureka Labs founder)
Tri Dao	0.53	reference_pilot (FlashAttention; Princeton faculty)
Tianshou	0.42	reference_pilot (RL library)
Jiayi Weng	0.33	reference_pilot (Tianshou author; OpenAI infra)
Suresh Kumar (Walmart CTO)	0.27	reference_pilot (sub-top-50 CTO)
tuixue.online	0.25	reference_pilot (Chinese visa-monitoring site)
Bojie Li	0.09	reference_pilot (this paper's author)
GPT-5 system-card-author mean	0.042	gpt5_system_card_author cohort

No prior instrument supports this placement. Citation count does not measure Altman, Srinivas, or Suresh Kumar; h-index does not measure the creator of nanoGPT; GitHub stars do not measure Yuval Harari; book sales do not measure Tri Dao. NameRank does measure them all, on the same scale.

6.2 The Credential Treadmill Thesis

Nine prestigious credentials, ranked by mean NameRank with the long-tail-OpenAlex-researcher cohort as baseline (Table 3):

Table 3: Credential ladder. Long-tail OpenAlex researchers ($n = 771$, citations 500–30,000) supply the baseline. Most prestigious-credential cohorts sit *at or below* the working-researcher baseline.

Credential	n	Mean NameRank	vs. baseline	vs. Yuval Harari
ICPC World Finalist gold	18	0.610	+0.146	−0.390
Putnam top-25 fellow	35	0.608	+0.144	−0.392
Long-tail OpenAlex (baseline)	**771**	**0.464**	—	−0.536
NOI China gold	29	0.368	−0.096	−0.632
IMO gold (2005–2015)	**197**	**0.362**	**−0.102**	−0.638
MSRA PhD Fellowship	237	0.343	−0.121	−0.657
IOI gold	68	0.341	−0.123	−0.659
Rhodes Scholarship	36	0.315	−0.149	−0.685
CMO China gold	131	0.302	−0.162	−0.698
CPhO China first prize	50	0.182	−0.282	−0.818
DeepSeek-V3 paper author	69	0.178	−0.286	−0.822
GPT-5 system card author	80	0.042	−0.422	−0.958

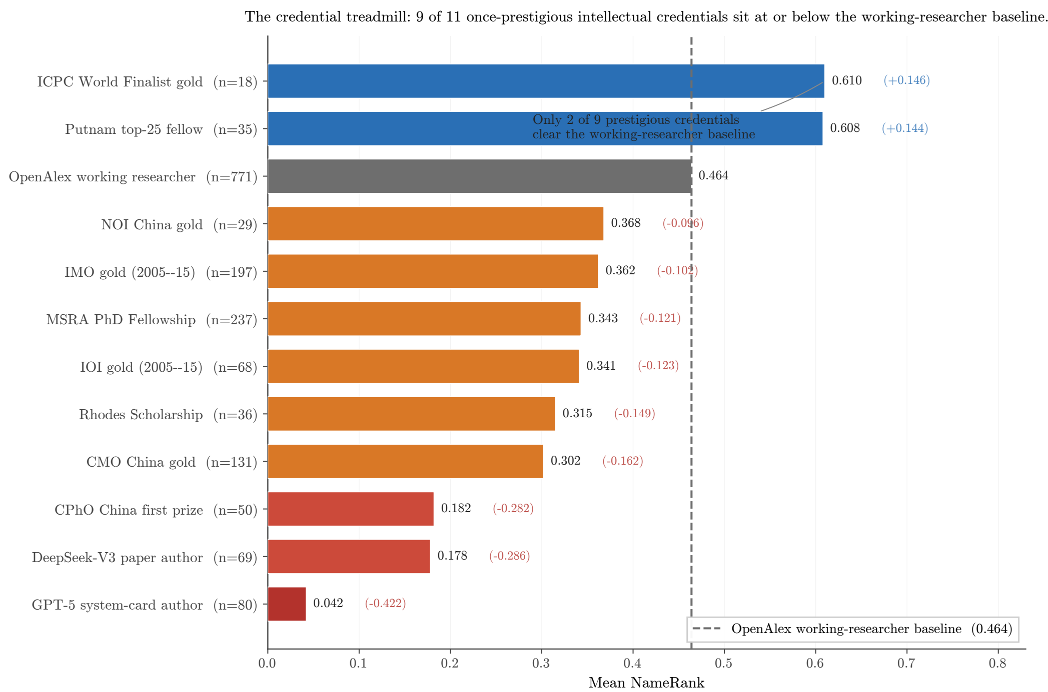

Figure 4: The credential treadmill (graphical form of Table 3). Bars are cohort mean NameRank; the dashed grey line is the long-tail-OpenAlex working-researcher baseline (0.464). Red bars: silent or near-silent cohorts. Orange: at or just below the baseline. Blue: above the baseline. Of nine prestigious-credential cohorts, only two (ICPC, Putnam) clear the working-researcher baseline.

The pattern is robust (Figure 4, Table 3). Seven of nine credentialed cohorts sit *below* the long-tail-OpenAlex working-researcher baseline. IMO gold ($n = 197$, the canonical intellectual credential of the past four decades) has mean NameRank 0.362, lower than the average citation-tracked researcher (0.464). Rhodes Scholars ($n = 36$) sit at 0.315, MSRA PhD Fellowship holders ($n = 237$) at 0.343.

Why ICPC and Putnam are exceptions. Both feed into US-tracked academic/industry careers with high English-corpus density. ICPC World Finalists ($n = 18$) become competitive programmers at FAANG companies and quant funds, with English-language profiles propagating well; Putnam top-25 fellows ($n = 35$) typically pursue US PhD programs in mathematics. These are not credential exceptions; they are proxies for selection into high-English-text-density career tracks.

Why IMO/CMO are not exceptions despite identical intellectual selection. IMO and CMO golds disperse globally, often becoming non-English-speaking researchers, mathematicians or engineers whose post-medal career produces text in languages other than English (German, Russian, Chinese, Korean, Japanese, Romanian, Farsi, etc.). The medal-and-name pair never co-occurs at high density in English training corpora. Conditioning on country reveals the gradient: IMO golds from US-tracked careers score ∼0.50; from non-Anglophone careers ∼0.30.

MSRA Fellowship temporal pattern. Within MSRA fellows, 5-year-cohort means rise from 0.312 (2005–2009) to 0.337 (2010–2014) to 0.382 (2015–2019) and dip to 0.343 (2020–2024)—consistent with the more recent fellows having had less time for post-fellowship career production to propagate. Pearson(year, NameRank) within IMO 2005–2015 is −0.042, indistinguishable from zero: the credential signal is not strengthening or decaying with time within the 10-year window.

Generational claim. Olympic-style credential systems were optimized for a pre-LLM signaling regime in which credentials translated into reputation through human social networks. In the LLM-mediated regime, the signal channel has shifted: only credentials that translate into *named, citable productions*—papers with the holder's name attached, OSS projects, blog posts, podcasts—propagate. The credential itself, absent subsequent production, does not.

6.3 The Named-Artifact-Amplification Mechanism

6.3.1 Artifact > creator inversion

Table 4 shows 11 verified (creator, artifact) pairs:

Table 4: Artifact > creator inversion. Columns: NR_c = creator NameRank, NR_a = artifact NameRank, $C \to A$ = fraction of model responses about the creator that mention the artifact, $A \to C$ = reverse. Boldface: the artifact's NameRank exceeds its creator's.

Creator	NR_c	Artifact	NR_a	$\Delta(a-c)$	$C \to A$	$A \to C$
Simon Willison	0.594	**Datasette**	**0.899**	**+0.305**	0.54	0.89
Dario Amodei	0.584	**Anthropic**	**0.811**	**+0.228**	1.00	0.41
Andrej Karpathy	0.614	**nanoGPT**	**0.707**	**+0.093**	0.05	0.76
Jiayi Weng	0.331	**Tianshou**	**0.420**	**+0.089**	0.25	0.00
Tri Dao	0.532	**FlashAttention**	**0.607**	**+0.075**	0.74	0.54
Lilian Weng	0.590	**lilianweng.github.io**	**0.614**	**+0.024**	0.97	0.97
Harrison Chase	0.479	**LangChain**	**0.502**	**+0.024**	1.00	0.22
Aman Sanger	0.289	**Cursor**	**0.305**	**+0.016**	1.00	0.00
Demis Hassabis	0.663	Google DeepMind	0.490	−0.173	1.00	0.62
Mira Murati	0.442	Thinking Machines Lab	0.222	−0.220	0.97	1.00
Aravind Srinivas	0.663	Perplexity	0.193	−0.470	1.00	0.33

In 8 of 11 pairs, the artifact's NameRank exceeds the creator's (Figure 5). The three exceptions are senior leaders of large named-product organizations (Hassabis/DeepMind, Murati/Thinking Machines, Srinivas/Perplexity), where the leader has substantial independent corpus presence from earlier credentials and the new organization is too young to have built equivalent name-attribution mass. For independent creators of single artifacts (Datasette, nanoGPT, Tianshou, LangChain, Cursor, FlashAttention, lilianweng.github.io), *the artifact uniformly exceeds the creator.*

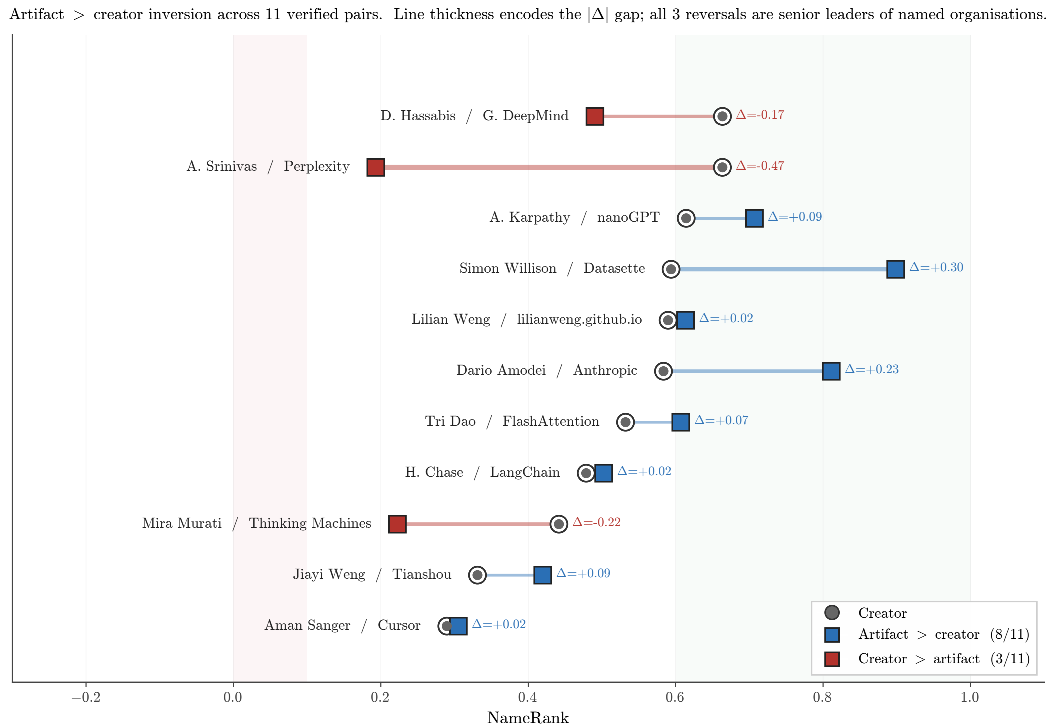

Figure 5: Artifact > creator inversion (graphical form of Table 4). Each row is a verified (creator, artifact) pair. Circle: creator NameRank; square: artifact NameRank; connecting line: the inversion direction. Blue squares (right of circle) indicate the artifact exceeds the creator (8 of 11 pairs). Red squares (left of circle) indicate the creator exceeds the artifact (3 of 11 pairs, all senior leaders of named organizations).

6.3.2 Conditional-probe asymmetry

The same table's last two columns quantify the direction of attribution flow. $C \to A$ is the fraction of model responses to "tell me about [creator]" that name the artifact. $A \to C$ is the reverse. Per-pair, per-model mention breakdowns are tabulated in Appendix D. Two regimes are visible:

- **Famous artifact, anonymous creator** ($C \to A$ high, $A \to C$ low): Aman Sanger / Cursor (+1.000 asymmetry), Harrison Chase / LangChain (+0.78), Aravind Srinivas / Perplexity (+0.67), Dario Amodei / Anthropic (+0.59). When probed about the creator, models reliably name the artifact; when probed about the artifact, models often fail to name the creator.

- **Paired in corpus** ($C \to A$ and $A \to C$ both high): Lilian Weng / lilianweng.github.io (0.97/0.97), Mira Murati / Thinking Machines Lab (0.97/1.00), Tri Dao / FlashAttention (0.74/0.54). The academic-norm pattern: creator and artifact co-occur reliably in both directions.

- **Inverse asymmetry**: Andrej Karpathy / nanoGPT ($C \to A = 0.05$, $A \to C = 0.76$). Karpathy is famous for too many things (Tesla, OpenAI, Stanford lectures); a probe about him rarely surfaces nanoGPT specifically, while probes about nanoGPT reliably surface Karpathy as the originator. This is structurally distinct from the famous-artifact-anonymous-creator regime.

6.3.3 Per-response mediation

On the Jiayi Weng probe specifically, mentioning Tianshou directly mediates the score:

Model class	n	Mean score
Models that mention “Tianshou”	7	**0.71**
Models that do not mention “Tianshou”	21	**0.35**

The seven mentioning models are the frontier-reasoning subset: claude-opus-4.6-think, claude-sonnet-4.6-think, gemini-3-flash-think, gemini-3.1-pro, glm-5.1-think, gpt-5.5-think, kimi-k2.6-think. The +0.36 score lift from artifact-mention is the empirical signature of the named-artifact-amplification mechanism at the per-response level—not inferred from cross-cohort patterns, but observed on a single entity across the panel.

6.4 External Validity

6.4.1 h-index dominates raw citations

On the long-tail-OpenAlex-researcher cohort ($n = 771$, citations 500–30,000, with h-index from OpenAlex):

Predictor	R^2 (within cohort)
log(citations)	0.137
log(h-index)	**0.398**
log(citations) + log(h-index)	0.398

Joint regression coefficients: log(h-index) +0.138, log(citations) −0.002. **Citations add zero marginal predictive power beyond h-index.**

Out-of-sample validation (70/30 train/test split on the same cohort): in-sample $R^2 = 0.344$, out-of-sample $R^2 = 0.498$, Pearson correlation 0.719. The relationship generalizes (and improves on held-out data, consistent with small-sample shrinkage). Full regression and per-decile breakdown are in Appendix F.

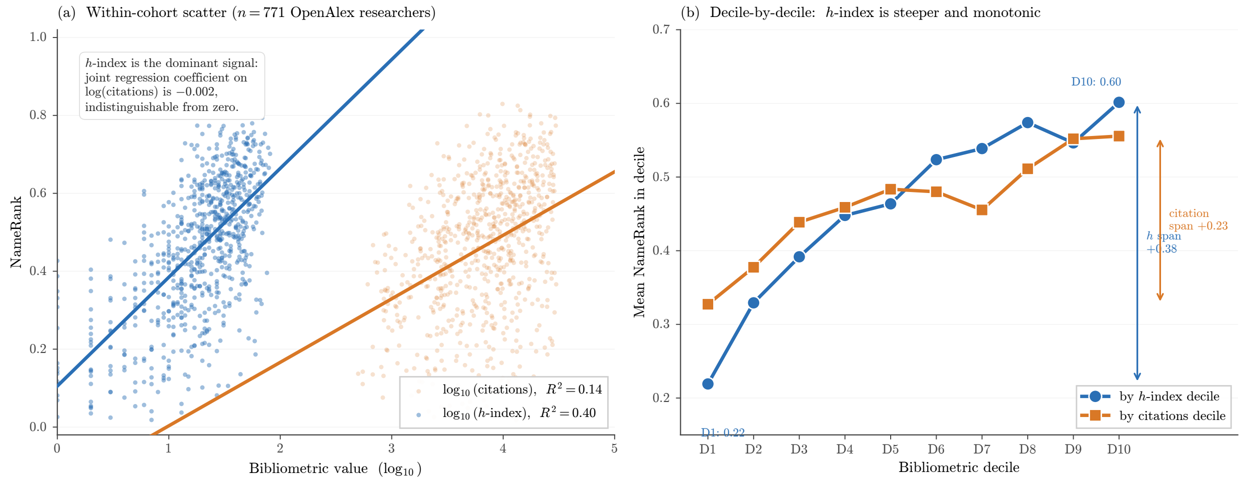

Figure 6: External validity: h-index dominates raw citations as a NameRank predictor. **(a)** Scatter of NameRank against $\log_{10}(h\text{-index})$ (blue) and $\log_{10}(\text{citations})$ (red) for the $n = 771$ OpenAlex long-tail-researcher cohort. The h-index regression has a steeper slope and explains $R^2 = 0.40$ of variance; the citation regression explains only $R^2 = 0.14$. **(b)** Mean NameRank by decile of each bibliometric: the h-index ladder is monotonic from 0.22 (D1) to 0.60 (D10), while citations show a much flatter 0.33-to-0.56 slope. In joint regression, citations contribute zero marginal R^2 beyond h-index.

Mechanistic interpretation. h-index measures the *distribution* of named-paper output—a sustained pattern of citable artifacts each carrying the author's name. Raw citation count can be dominated by 1–2 viral papers with hundreds of co-authors and weak per-author name attribution (the hyperauthorship problem [Cronin, 2001]). NameRank is a *named-attribution metric*, so h-index is the right bibliometric feature; citation volume is the wrong one.

IKP Sec. 5.7 replication, refined. Li [2026] reports that bibliometrics explain ∼35% of recognition variance, with citations as the bibliometric. With h-index instead, we explain 40%. The original ∼35% figure was an average across these features; the right decomposition is that the h-index channel carries the bibliometric signal and raw-citation channel carries nearly none of it.

6.4.2 Institution and country gradient

Within the CS faculty cohort ($n = 698$ across 320 distinct institutions), institution-as-categorical-predictor alone explains 54.4% of NameRank variance (Table 5).

Table 5: CS faculty NameRank by institution (15 selected institutions with $n \geq 4$). Stanford-affiliated faculty score 2× Tsinghua-affiliated. The Tsinghua/Peking samples are small ($n = 4$ each); the gradient holds across larger samples (CMU, Cornell, Michigan, Princeton, etc.).

Institution	n	Mean	Min	Max
Stanford University	19	**0.537**	0.14	0.62
University of Washington	26	0.529	0.13	0.68
Cornell University	33	0.484	0.16	0.67
Carnegie Mellon University	65	0.484	0.09	0.66
Princeton University	25	0.443	0.18	0.61
HKUST	5	0.462	0.24	0.58
Johns Hopkins University	4	0.453	0.19	0.67
TU Delft	5	0.445	0.30	0.58
University of Michigan	31	0.429	0.16	0.67
Universidade de Lisboa	5	0.427	0.26	0.52
Georgia Institute of Technology	6	0.462	0.23	0.65
Peking University	4	0.290	0.22	0.34
Tsinghua University	4	**0.258**	0.15	0.35
USTC	4	0.310	0.11	0.45
Monash University	6	0.386	0.27	0.65

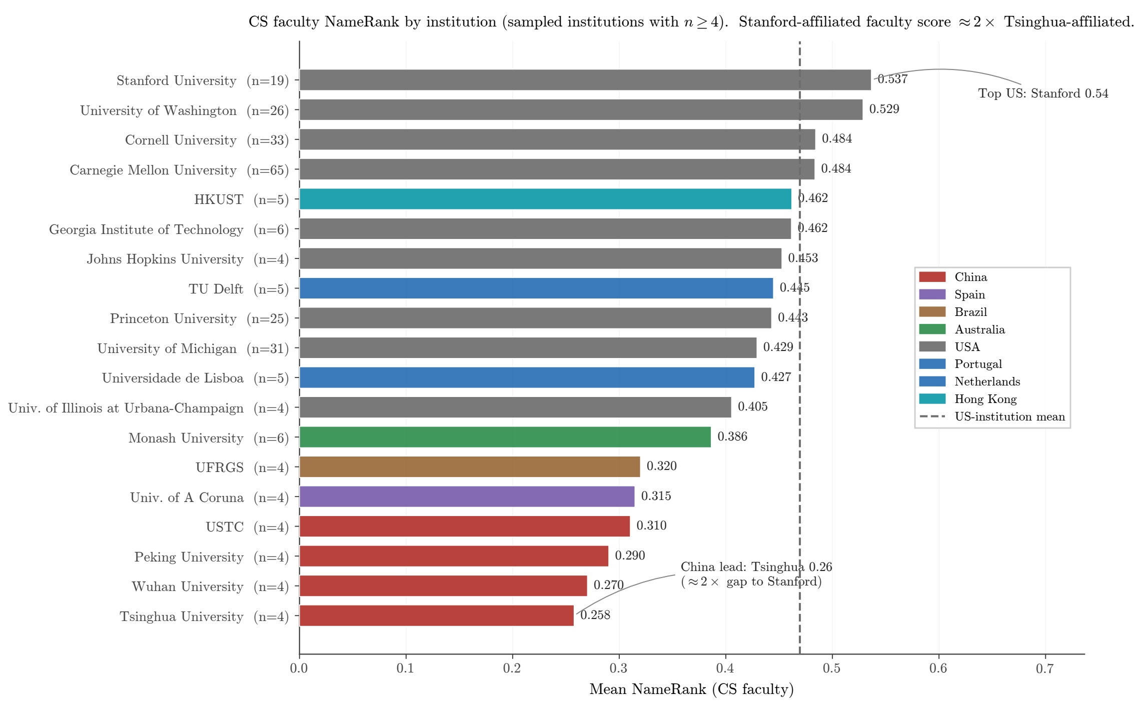

Figure 7: CS faculty NameRank by institution (graphical form of Table 5). Bars are colored by country; the dashed line marks the mean of US-institution bars. The Stanford–Tsinghua gap exceeds 2×. Chinese institutions cluster at the bottom of the chart even though Tsinghua, Peking, and USTC have global research reputations on par with the upper US cluster—the gradient is corpus-density, not research quality (Section 6.4.2).

Country-level gradient. Aggregating institutions by country (Table 6, Figure 8):

Table 6: CS faculty NameRank by country of institutional affiliation. Countries with ≥ 3 sampled faculty; `Other/Unknown` (n=236) consists of CS-Rankings affiliations not matched by the keyword-prefix country lookup and is omitted from the gradient discussion.

Country	n	Mean	Median	frac ≥ 0.5
Israel	7	0.506	0.528	57%
Singapore	7	0.497	0.480	43%
Sweden	5	0.492	0.490	40%
Germany	9	0.490	0.488	44%
Switzerland	4	0.489	0.467	50%
Canada	9	0.478	0.463	22%
USA	302	0.460	0.488	46%
Netherlands	9	0.447	0.432	33%
UK	16	0.438	0.440	38%
Hong Kong	11	0.431	0.452	36%
Portugal	5	0.427	0.442	20%
Brazil	10	0.388	0.409	10%
Australia	11	0.341	0.291	18%
France	3	0.312	0.341	0%
China	30	**0.311**	0.285	7%
Spain	8	0.303	0.288	13%
India	10	0.292	0.279	0%

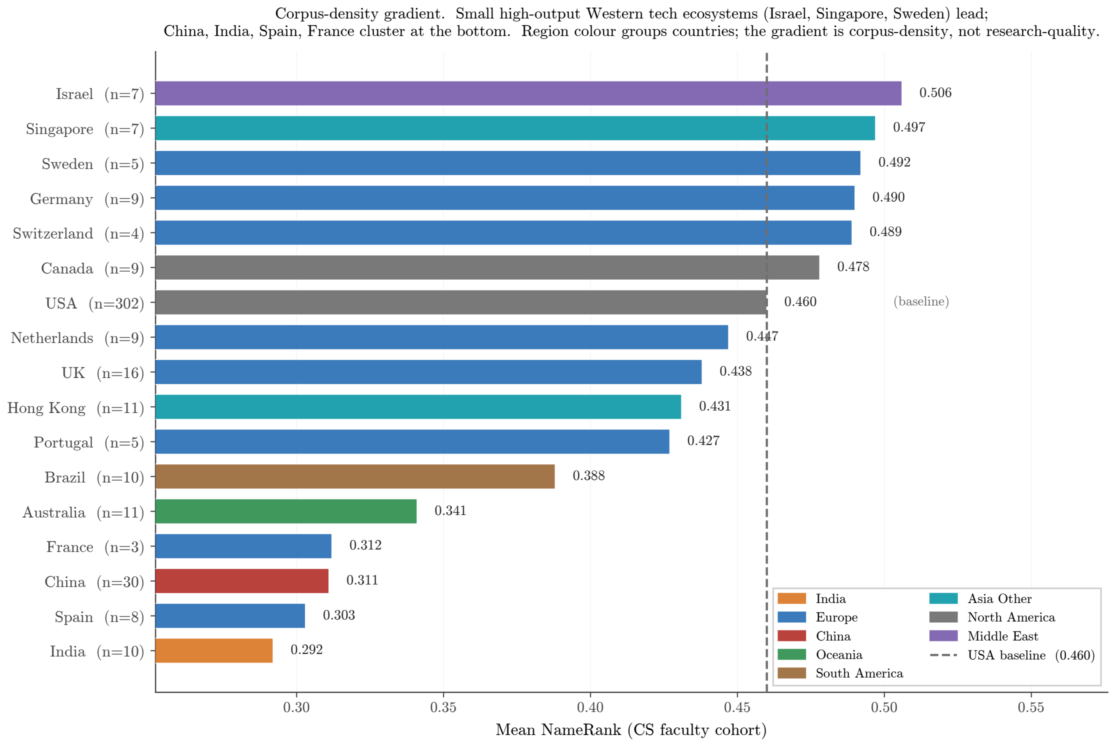

Figure 8: Country gradient in CS faculty NameRank. Small high-output Western tech ecosystems (Israel, Singapore, Sweden) score above the USA baseline (dashed); Chinese, Indian, French, and Spanish CS faculty cluster at the bottom. The gradient is corpus-density, not research-quality: Tsinghua, Peking, USTC and similar institutions have global research reputations comparable to upper US institutions but produce far less English-language web text per affiliation. France ($n = 3$) and Switzerland ($n = 4$) samples are small.

The gradient is corpus-density, not research-quality. Tsinghua is China's top CS school; the gap to Stanford is not a quality gap. It is a measure of English-language web-text production volume per affiliation. Small high-output Western tech ecosystems (Israel, Singapore, Sweden) lead the gradient, the USA sits in the middle, and Chinese, Indian, French CS faculty cluster at the bottom—even though some Chinese institutions (Tsinghua, Peking, USTC) have global research reputations on par with the upper US cluster.

The Tsinghua bind. A young Chinese CS faculty member at Tsinghua produces papers, lab pages, and conference talks that are equally rigorous as their CMU counterpart but propagate primarily through Chinese-language channels (Zhihu, WeChat technical articles, Chinese-language tutorials) plus their English papers. The CMU counterpart's same output additionally propagates through Twitter/X, Substack, English-language podcast appearances, US-news coverage of the institution, and a denser tutorial ecosystem. The CMU researcher's name-attribution mass per unit of research output is structurally higher. NameRank measures the result.

6.5 East/West Model Variance and Cross-Language Sub-Run

6.5.1 English-prompt East-West signal

Partitioning the 37-model panel into 23 Western and 14 Chinese models, the cohort-level median Chinese-minus-Western delta is *positive across nearly every cohort*, ranging from +0.02 to +0.10.

The Chinese-minus-Western lift does not reflect Chinese models preferentially recognizing Chinese-content entities; rather, Chinese models commit globally to confident-but-accurate-enough descriptions where Western models more readily refuse on unfamiliar names. Cohorts with the largest positive C–W medians: ai_hardware (+0.096), mid_tier_architect (+0.067), research_paper (+0.062), ai_startup_or_company (+0.062), mid_tier_gov_ai_policy (+0.062).

Individual cases reveal more structure. Strongest Chinese-content lift: Wei Dongyi (Chinese math figure) C = 0.80 vs. W = 0.53 (+0.27); Zhang Fang (NOI China gold) C = 0.45 vs. W = 0.20 (+0.25). Strongest Western lift: Saad Albawi (Arab-named Western researcher) W = 0.59 vs. C = 0.16 (−0.42, Chinese models refuse on the name); Qingzi Vera Liao (Chinese-name CS faculty at Western institution) W = 0.50 vs. C = 0.15 (−0.35).

Interpretation. The Chinese-content lift is real but small (median +0.03–0.10 at cohort level), while the willingness-to-commit gap is large: the high-refusal cluster (gpt-oss-20b-think at 48%, grok-4.20-think at 46%, mistral-small-24b at 46%, qwen3-235b-a22b-think at 42%) refuses heavily and contains both Western and Chinese models; the zero-refusal cluster (gemma-3-4b, phi-4, ernie-4.5-300b-a47b at 0%, llama-3.2-1b/ministral-3b/llama-4-maverick at $\leq 0.1\%$) commits regardless and also spans both classes. The dominant signal is *threshold-of-confidence* asymmetry between training regimes (RL-tuned-against-hallucination vs. fluent-completion-by-default), not corpus-coverage asymmetry between Western and Chinese training data.

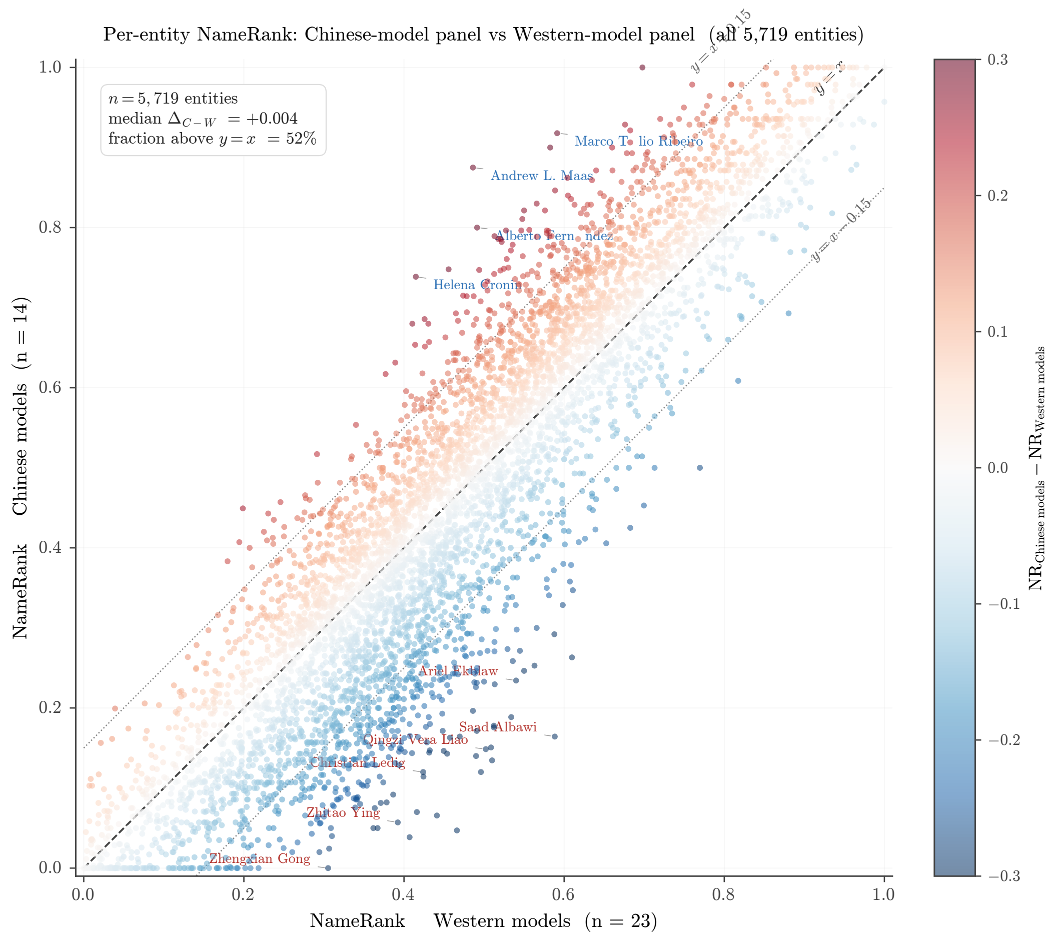

Figure 9: Per-entity NameRank from the Chinese-model panel ($n = 14$) plotted against the Western-model panel ($n = 23$) across all 5,719 entities. Color encodes $\Delta = \mathrm{NR_C} - \mathrm{NR_W}$ (clipped at ± 0.30 for legibility). The cloud follows $y = x$ with a faint upward tilt (52% of entities sit above the diagonal, median $\Delta = +0.004$). The most divergent named entities cluster along the off-diagonals: Chinese-name entities tend to sit above the line; English-coded entities tend to sit below. The dominant signal is per-entity, not per-model-class.

6.5.2 Chinese-prompt sub-run

We re-probed 240 entities with the Chinese-translated probe template (gold answer left in English; judge handles cross-language). The headline finding is the *prompt-language* $\times$ *entity-name interaction* (Table 7):

Table 7: Cross-language NameRank: $\Delta(\mathrm{NR_{zh}} - \mathrm{NR_{en}})$ by cohort. Chinese-prompt lifts Chinese-name entities and slightly suppresses English-coded entities.

Cohort	n	$\mathrm{NR_{en}}$	$\mathrm{NR_{zh}}$	Δ
deepseek_v3_author	69	0.178	**0.247**	**+0.069**
mid_tier_filmmaker	5	0.765	0.802	+0.037
noi_china_gold	29	0.368	0.392	+0.025
cmo_china_gold	30	0.301	0.315	+0.013
cpho_china_first_prize	30	0.184	0.194	+0.010
mid_tier_politician	5	0.361	0.364	+0.003
mid_tier_writer	5	0.650	0.631	−0.020
msra_phd_fellowship	30	0.339	0.319	−0.021
reference_pilot	37	0.452	0.422	**−0.030**

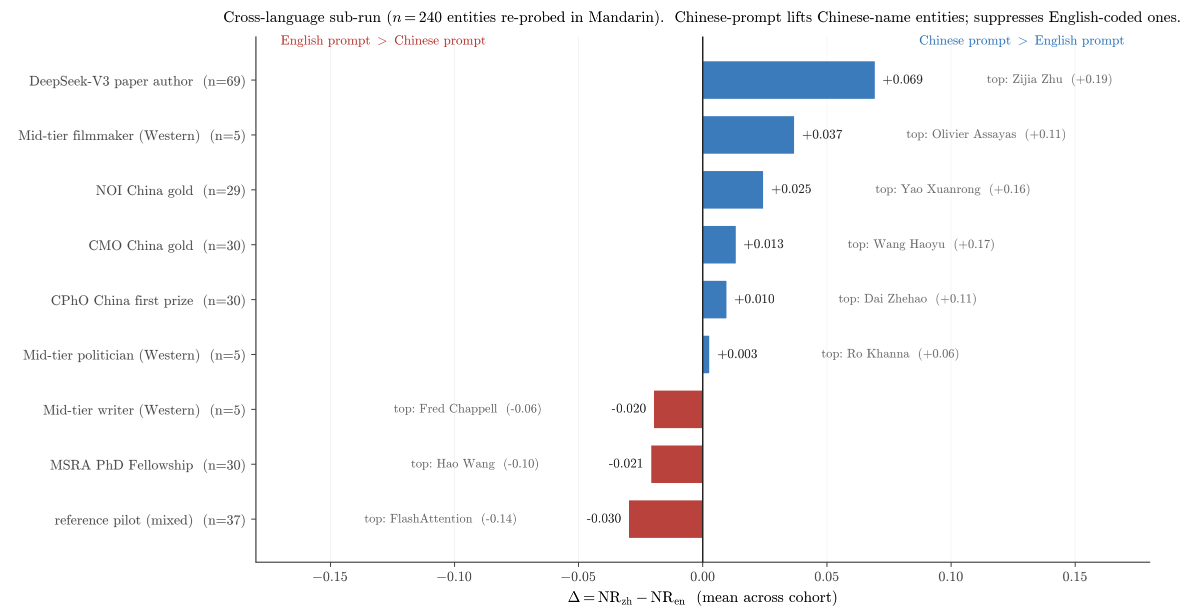

Figure 10: Cross-language sub-run (graphical form of Table 7). Bars are cohort means of $\Delta = \mathrm{NR_{zh}} - \mathrm{NR_{en}}$; the labeled italic name to the side of each bar is the cohort's strongest signed exemplar. Chinese-prompt lifts Chinese-character-name cohorts (DeepSeek-V3 authors at the top, NOI/CMO/CPhO China cohorts) and suppresses English-coded ones (reference_pilot exemplar: FlashAttention at −0.14).

Per-entity extremes. Largest Chinese-prompt lifts: Wang Haoyu (+0.17), Yao Xuanrong (+0.16), Wang Zhongjian (+0.16), Zijia Zhu (DeepSeek author, +0.19). Largest Chinese-prompt suppressions: FlashAttention (−0.14), Aleksander Madry (−0.12), Du Zhuofan (−0.15), Leng Junsheng (−0.17). The signature is consistent: Chinese-character names gain on Chinese prompts; English-coded artifacts (FlashAttention, ReAct) and English-coded individuals (Madry) lose.

Per-model deltas: language effect is universal, not Chinese-model-specific. Models that gain on Chinese prompts include both Western (claude-sonnet-think +0.17, claude-opus-think +0.11, gemini-3.1-pro +0.12, gemma-3-12b +0.12) and Chinese (deepseek-v4-pro-think +0.14, qwen3-235b +0.10). Models that lose include Western (llama-4-maverick −0.20, gpt-5.4 −0.12, mistral-large −0.09) and Chinese (glm-4-32b −0.05, kimi-k2 −0.05). Average Western model:

+0.014; average Chinese model: +0.028. Both classes lift slightly; the dominant signal is entity-name match, not model class.

Corpus-pocket activation as the mechanism. Even Western models trained on multilingual data have *Chinese-text pockets* about Chinese-named entities; a Chinese prompt unlocks those pockets. Inversely, an English-coded artifact (FlashAttention) is best recalled from English text; Chinese probes route retrieval to less-dense Chinese text about it. The corpus is internally structured by language; prompt language is the routing signal.

Methodological implication. NameRank should be measured in the dominant-corpus language of the entity for fair within-entity scoring. Cross-entity comparisons should be done within consistent prompting language to control for the routing effect. Our headline results use English prompts; the Chinese-prompt scores for the 240-entity subset are released as a separate dataset, with selected per-entity deltas in Appendix E.

6.6 Case Studies: C5 and C6

6.6.1 C5—Jiayi Weng and named-artifact amplification

Jiayi Weng (Tianshou author, OpenAI infrastructure engineer) was the originator of the life-score = number-of-people-who-remember-your-name formulation that motivates this paper. We measure him against eight broadly-matched OpenAI engineering/research individual contributors from publicly-listed model authorship attributions: Aaditya Singh, Adam Fry, AJ Ostrow, Alex Wei, Andrea Vallone, Pamela Vagata, Trevor Blackwell, and Vicki Cheung. We label the comparison set "matched" in the loose sense that all are OpenAI ICs with public technical-contribution attribution; we do not claim formal matching on hiring year, role title, or pre-OpenAI background. The point of the comparison is to show that Jiayi's NameRank lift over the comparison set is driven by a named, attributable, well-named artifact (Tianshou) rather than by the OpenAI affiliation that everyone in the set shares.

Entity	NameRank	Refusal rate
Jiayi Weng	**0.331**	24%
Tianshou (artifact)	0.420	3%
Trevor Blackwell	0.431	8%
Pamela Vagata	0.184	32%
Vicki Cheung	0.300	24%
Aaditya Singh	0.027	68%
Adam Fry	0.084	51%
AJ Ostrow	0.078	60%
Alex Wei	0.039	51%
Andrea Vallone	0.076	57%
Matched-peer mean	0.152	—-
Matched-peer range	[0.027, 0.431]	—-
Jiayi Weng z-score within peers	**$+1.24\sigma$**	—-

The $+1.24\sigma$ lift over the comparison set is artifact-mediated: Tianshou's NameRank (0.420) is *higher* than its creator's (0.331). The mechanism is the named-artifact-amplification mechanism documented at scale in Section 6.3. Both English and Chinese prompts produce essentially identical NameRank for Jiayi (0.331 English, 0.327 Chinese) and Tianshou (0.420 English, 0.399 Chinese), confirming the attribution channel is language-invariant for this case.

6.6.2 C6—tuixue.online falsification

tuixue.online is a Chinese-language website monitoring US F1 visa appointment availability for international applicants, maintained between 2019–2024 with a peak user base of ∼1M Chinese F1 applicants who relied on it during the post-COVID visa backlog. By any conventional metric of human reach (active users, monthly visitors, downstream effects on visa applications received and processed), tuixue.online's reach substantially exceeds that of nanoGPT (whose audience is order-of-magnitude tens of thousands of developers).

Entity	NameRank	Approx. human reach
tuixue.online	**0.250**	**∼1M Chinese F1 visa applicants**
nanoGPT	0.707	∼10K developers
FlashAttention	0.607	academic + library users
ReAct (paper)	0.578	academic
vLLM	0.527	∼100K developers
LangChain	0.502	∼100K developers

NameRank for tuixue.online is less than half of nanoGPT's despite vastly larger human reach. The cross-language null—$\mathrm{NR}_{\mathrm{en}} = 0.250$, $\mathrm{NR}_{\mathrm{zh}} = 0.262$, $\Delta = +0.012$—confirms that the dissociation is not an English-corpus artifact. Both English and Chinese training corpora lack the URL-creator co-attribution that would propagate the artifact's name. The corpus-presence absence is symmetric.

Why. tuixue.online was discussed in Chinese-language WeChat groups and on Zhihu, but the discussion typically referenced the URL functionally (*"tuixue.online has visa appointment slots open for India"*) without attaching the creator's name to the URL in indexable form. Open-source projects with comparable utility (nanoGPT, vLLM, LangChain) have creators publicly identified on README files, conference talks, paper bylines, and tutorial videos—attaching their names to the artifact name across many indexable documents.

Falsification interpretation. C6 establishes the boundary condition for NameRank: it measures *indexed corpus presence of the entity's name*, not human utility or social reach. The two correlate for academic/OSS artifacts because creators attach their names to their work. They dissociate for utility-driven artifacts that propagate by word-of-mouth or through non-English channels with anonymous-by-convention attribution. The dissociation is not a flaw to be patched; it is what the metric measures. C6 is the sharpest case demonstrating that NameRank measures something specific and falsifiable.

6.7 Methodological Validation

6.7.1 Variance decomposition

For 211,603 records across 5,719 entities × 37 models × 54 cohorts, we report each factor's independent share of the total sum of squares. Entity and Cohort are nested (each entity belongs to exactly one cohort), so their SS contributions overlap and the rows do not partition the total; each row gives the fraction of total SS that factor independently explains.

Source	SS	% of total
Entity	9,417	**35.8%**
Model	2,873	10.9%
Cohort	4,425	16.8%
Residual (interactions)	14,035	53.3%

$\sigma^2_{\text{entity}}/\sigma^2_{\text{model}} = 3.3$: the entity signal dominates model-level bias by more than a factor of three. NameRank measures the entity, not the panel. Per-cohort within-entity disagreement is broken down in Appendix G.

Per-model marginal mean scores (a proxy for model generosity):

Generous models (mean > 0.55)	
gemini-3-flash-think	0.660
gpt-5.5-think	0.619
deepseek-v4-pro-think	0.604
claude-opus-4.6-think	0.603
ernie-4.5-300b-a47b	0.603
Strict models (mean < 0.30)	
qwen3-32b-think	0.284
qwen3-8b-think	0.286
gpt-oss-20b-think	0.265
mistral-small-24b	0.225
llama-3.2-1b	0.201

Thinking mode does not correlate systematically with generosity (both gemini-3-flash-think and qwen3-32b-think are thinking-mode). The strict cluster is dominated by smaller models and by RL-tuned-against-hallucination defaults (gpt-oss, qwen3, mistral-small); the generous cluster has the frontier reasoning models. Full per-model generosity and refusal statistics for all 37 models are in Appendix C.

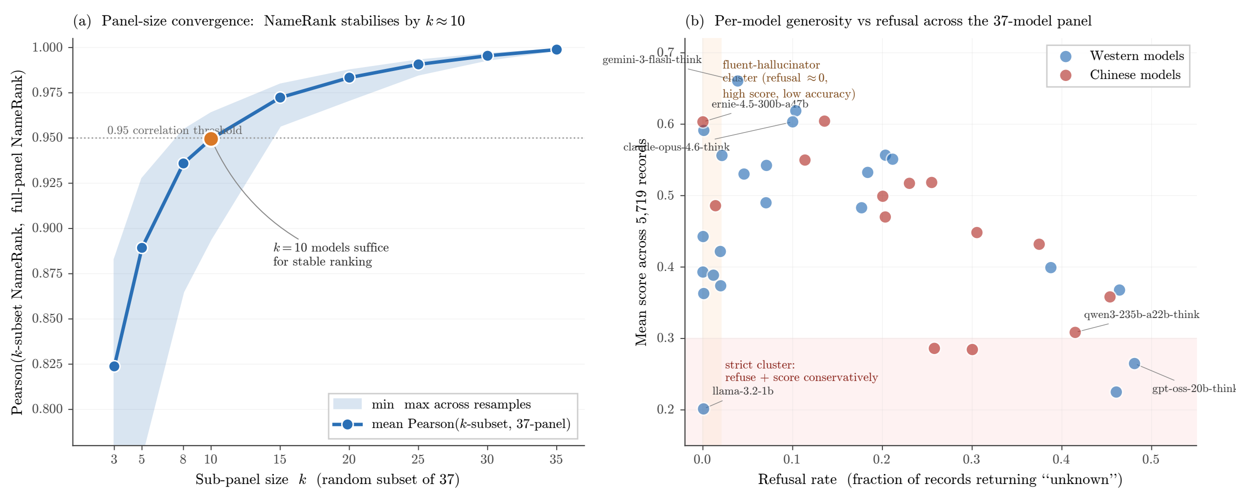

Figure 11: Panel-size and per-model robustness. **(a)** Pearson correlation between NameRank computed from a random k-subset of the panel and NameRank from the full 37-model panel, averaged over 50 resamples per k; the shaded envelope shows min/max across resamples. The metric stabilizes at $r \geq 0.95$ by $k \approx 10$; subsequent models contribute steadily-diminishing marginal information. **(b)** Per-model generosity (mean score across 5,719 entities) versus refusal rate, colored by West/East partition. The two failure regions are visible: a strict cluster in the lower right (RL-tuned-against-hallucination models that refuse and score conservatively) and a fluent-hallucinator cluster on the left (refusal ≈ 0 with high score), both spanning Western and Chinese vendors.

6.7.2 Refusal patterns

The refusal rate (fraction of panel responding “unknown”) inverts NameRank almost perfectly at cohort level:

Cohort	Refusal rate	NameRank mean
gpt5_system_card_author	**58%**	**0.042**
mid_tier_yc_company	57%	0.101
cpho_china_first_prize	53%	0.182
long_tail_researcher_ikp	47%	0.158
deepseek_v3_author	42%	0.178
...	...	...
research_paper	3%	0.625
mid_tier_book	3%	0.363
oss_project	3%	0.519

Models honestly refuse on low-NameRank cohorts rather than hallucinating, which is the threshold-cleanliness property surfacing in raw response behavior.

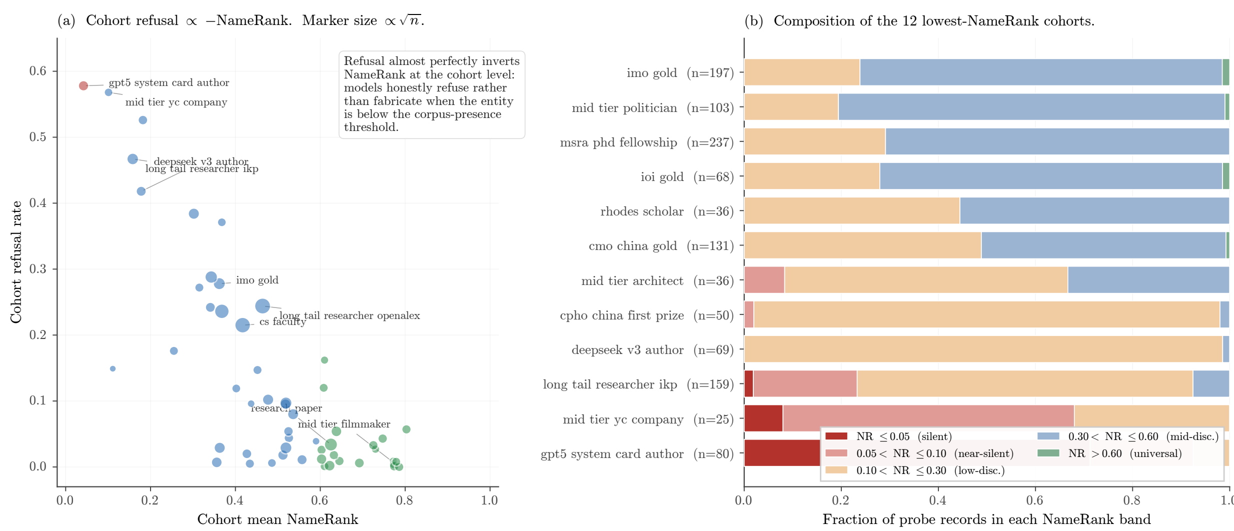

Figure 12: Refusal and silent-zone composition. **(a)** Cohort mean NameRank versus cohort refusal rate; marker size scales with $\sqrt{n}$. The inverse relationship is near-monotone: cohorts with low NameRank are dominated by refusals, not hallucinations. **(b)** Per-record composition of the 12 lowest-NameRank cohorts in terms of the NameRank band each record falls into. The silent band (NR $\leq$ 0.05, dark red) is heavy at the bottom of the chart and tapers smoothly with mean NameRank—the metric does not produce a noisy mid-band tail of fabricated mid-confidence records.

Per-model refusal signature. The very-high-refusal cluster (gpt-oss-20b 48%, grok-4.20-think 46%, mistral-small-24b 46%, minimax-m2.7-think 45%) is dominated by RL-tuned-against-hallucination defaults. The zero-refusal cluster (llama-3.2-1b 0.001%, gemma-3-4b 0%, phi-4 0%, ernie-4.5-300b 0%, ministral-3b 0.001%, llama-4-maverick 0.001%) consists of *fluent hallucinators*—models that always produce a confident-looking bio whose multiplicative coverage×accuracy correctly down-weights through the accuracy axis (→ 0 when the bio is wrong). This validates the multiplicative score: a coverage-only or refusal-only metric would systematically over-credit the fluent-hallucinator cluster.

6.7.3 Judge vs. embedding gap

Pearson(judge score, embedding similarity) = 0.152 across 175,397 non-refusal records. Embedding similarity is essentially flat across judge-score buckets:

Judge score bucket	n	Mean embedding_sim
0.0–0.1	23,621	0.819
0.1–0.3	14,358	0.798
0.3–0.5	26,291	0.801
0.5–0.7	29,616	0.818
0.7–0.9	52,534	0.843
0.9–1.0	28,977	0.829

Embedding measures topical/lexical similarity, which is high even for fluent-hallucination bios because the wrong-bio of a CS researcher still embeds close to the gold-about-that-researcher's actual subfield. Judge measures factual accuracy.

The most informative disagreement cases (judge low, embedding high):

Entity	Model	Judge	Emb.	Gap
Hao Tang	claude-opus-think	0.00	0.97	+0.97
Yuji Roh	phi-4	0.00	0.97	+0.97
Xuanhe Zhou	mistral-medium-3.1	0.00	0.97	+0.97
Linfeng Zhang	mistral-medium-3.1	0.00	0.96	+0.96
Daya Guo	mistral-medium-3.1	0.00	0.96	+0.96

All Chinese-name long-tail CS researchers. Models produce confident bios in the actual subfield—embedding rates 0.97—but the bios are factually about the wrong person or fabricated. The judge correctly assigns 0.

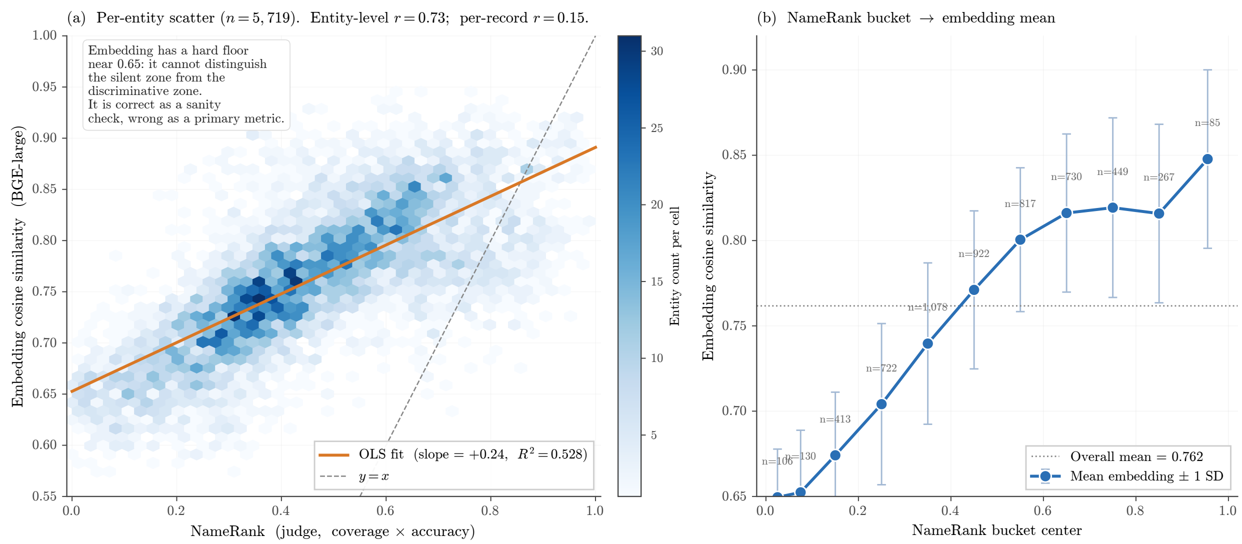

Figure 13: Embedding cross-check at the entity level. **(a)** Per-entity hexbin scatter of judge-derived NameRank against mean embedding similarity (n = 5,719 entities). Entity-level Pearson $r = 0.73$ understates the gap: the per-record correlation across the 175,397 non-refusal records is only $r = 0.15$. **(b)** Mean embedding similarity by NameRank bucket. The embedding floor is ≈ 0.65 even in the silent zone; the slope is shallow. An embedding-only metric cannot distinguish a fluent hallucination from a correct response, which is why NameRank uses the LLM judge as the primary instrument.

Implication. An embedding-only NameRank would systematically credit fluent hallucinations on Chinese-name long-tail researchers as "recognized" when models are in fact hallucinating about them. Multiplicative coverage×accuracy with an LLM judge is the necessary defense; we cannot substitute embedding similarity for the judge step.

7 Discussion

7.1 What We Have Built and What We Have Not

NameRank is a continuous cross-domain instrument with empirical threshold cleanliness, validated external correlates (h-index, institution), and mechanism-level structure (named-artifact amplification, conditional-probe asymmetry). It is suitable for: cross-domain placement of individuals and artifacts on a common scale; tracking propagation of specific names into frontier-model training corpora over time; measuring the recognition-impact channel that bibliometrics misses (the IKP Sec. 5.7 residual). It is not suitable for: quality assessment (it measures what models absorbed, not what is worth absorbing); attribution of authorship to specific contributions (it measures the name, not the deed); inference about future events (a 2026 measurement reflects 2025-and-earlier corpus state).

7.2 The Credential Treadmill Thesis in Context

Section 6.2 shows that IMO/IOI/CMO/Rhodes/MSRA credentialed populations sit at or below the long-tail-OpenAlex-researcher baseline in NameRank. This is not a claim that the credentials are unrelated to subsequent recognition; the literature documents strong causal effects of olympiad medals on subsequent academic productivity [Agarwal and Gaule, 2020, Lubinski et al., 2020]. Rather, it is a claim about the post-credential propagation channel. When the dependent variable is published-paper count or academic-prize attainment (as in the prior literature), credentials predict positive outcomes. When the dependent variable is name-propagation into LLM training corpora, credentials per se are not sufficient—continued production of named, indexable artifacts is.

The implication is methodologically subtle but consequential. A credentialing institution selecting candidates from a pool of Olympiad medalists is selecting on a signal that was informative in the pre-LLM regime (because credentialed candidates were observable through human social networks and bibliometric signals). In the LLM regime, the same selection rule selects candidates whose names will likely not propagate into frontier-model training data unless they additionally produce named artifacts with clean attribution. The credential is a near-zero predictor of post-credential NameRank trajectory; the named-artifact-attached career is the predictor.

We do not claim the credential is therefore useless—it remains informative about cognitive ability, conscientiousness, and field-specific motivation. We claim only that its role as a *name-propagation channel* has shrunk substantially in the LLM era.

7.3 The Artifact > Creator Inversion and the Anonymity-of-Tooling Problem

For independent creators of well-named single artifacts, the artifact uniformly exceeds the creator in NameRank (Section 6.3). The structural finding is that named tools propagate further than their makers. Cursor at NameRank 0.31 is more famous than Aman Sanger at 0.29. LangChain at 0.50 exceeds Harrison Chase at 0.48. nanoGPT at 0.71 exceeds Karpathy's 0.61 even though Karpathy is unusually prominent in his own right.

The mechanism is the asymmetry of attribution direction (Section 6.3). Probes about creators reliably surface the artifact, because creator-bios in indexable text are written to feature creations. Probes about artifacts often fail to surface the creator, because artifact-discussion text in OSS, tutorial, and documentation contexts mentions the artifact name far more often than the creator's. The structure of derivative content favors the tool over the toolmaker.

A practitioner consequence: for an individual seeking to propagate their name through the LLM channel, building a named tool is multiplicatively more effective than writing many papers—but the multiplier accrues to the tool's name first, and to the creator's name only through the tool's name. The creator who buries the tool's name behind corporate-product naming or generic branding collects no NameRank lift from creation. The implication for individual-researcher career strategy is uncomfortable but actionable: *name the tool clearly, attach the name to its README and to every paper that uses it.*

7.4 The Corpus-Density Gradient and Its Limits

Section 6.4.2 shows a robust Stanford= 0.54 vs. Tsinghua= 0.26 gap that scales by country (Israel/Singapore/Sweden lead; China/India/France lag). The mechanism is English-corpus density; the consequence is structural inequality in name-propagation per unit of research output.

Three caveats. First, the Tsinghua/Peking samples in our cohort are small ($n = 4$ each); the country-level gradient is well-sampled (China $n = 30$) but institution-level reads are noisy at the bottom. Second, the gradient may partly reflect researcher mobility: Chinese-born researchers who emigrate to US institutions score under their US institution, which boosts the US average and depresses the China average. We have not corrected for this. Third, the gradient is downstream of indexing and crawling decisions made by training-data pipeline operators; it could in principle be reduced by changing those decisions. Within the current corpus regime, however, it is a substantial structural effect.

7.5 Application to Cross-Domain Talent Assessment

The cross-domain unity property (Section 6.1) suggests an application that is structurally absent from the current toolkit: talent-based immigration adjudication (US EB-1A/O-1, UK Global Talent, Australian Distinguished Talent, and equivalents), where non-expert adjudicators must evaluate applicants spanning academia, industry, OSS, and the arts using non-interconvertible proxies (citation count, press coverage, awards, selective-organization membership). NameRank is the first instrument that places a Walmart CTO, a Stanford CS professor, an OSS maintainer, and a popular nonfiction author on the same axis. For criteria that explicitly evaluate *recognition*—the "sustained national or international acclaim" prong of EB-1A, the O-1B fame-in-the-field criterion—this is precisely the question the instrument answers.

The structural fit ends there. Five findings in this paper make NameRank unsuitable as a primary or weighted criterion in such adjudication, and the same findings should govern responsible use as a supplemental signal.

The corpus-density gradient is constitutive. Section 6.4.2 places Stanford at 0.54 and Tsinghua at 0.26 for reasons mechanistically about English-language web-text production per affiliation, not research quality. A talent-immigration program is, almost definitionally, identifying exceptional non-domestic candidates—often from precisely the lower-density-corpus geographies (China, India, France) that the gradient suppresses. Using NameRank as a criterion would systematically downscore the applicants the program exists to find.

The credential treadmill predicts directional disagreement with existing instruments. Seven of nine prestigious-credential cohorts (Section 6.2) sit below the long-tail-OpenAlex working-researcher baseline—and these are exactly the credentials USCIS and equivalent agencies accept as evidence of extraordinary ability. NameRank and the existing adjudication criteria will not just produce noisy correlations on this population; they will disagree directionally on the most prestigious applicants.

C6 is a worked example of the failure mode. tuixue.online ($\sim$1M users, Section 6.6) scores 0.250 against nanoGPT's 0.707 ($\sim$10K users). A high-impact builder operating in non-English channels with conventional anonymous-by-default attribution is invisible to the instrument. Applicants from non-Anglophone countries who built widely-used tools in their domestic language register as low-recognition despite producing exactly the kind of public-good output these programs are designed to reward.

Name-collision attenuation introduces ethnicity-mediated bias. Li [2026] measured $\sim 2\times$ recognition attenuation for common East Asian surnames in the IKP $2\times2\times2$ audit. NameRank inherits this property by construction. Whether the bias is descriptively justified as "what the corpus has absorbed" or not, it would propagate directly into any adjudication decision anchored to the metric.

Gaming risk under high-stakes use. An evaluation criterion with strong individual incentives attracts optimization. Tutorial-farming, sponsored-blog placement, commissioned Wikipedia coverage, and AI-generated derivative material all directly increase NameRank at modest cost. The credential-treadmill finding shows the market already knows which channels propagate; formalizing NameRank as a visa criterion accelerates substitution into those channels without changing the underlying talent distribution—a Goodhart's-law dynamic the descriptive measurement is otherwise insulated from because no one is currently optimizing against it.

Responsible use as a supplemental signal. A narrower legitimate role survives: a cross-domain sanity check that lets a non-expert adjudicator confirm a non-academic applicant has achieved recognition comparable to an academic baseline (a founder at 0.50 is comparable to a mid-career CS professor at the same value). Threshold cleanliness (Section 6.7) licenses this use—an applicant scoring near zero should be read as "not propagated into the indexed corpus," uninformative rather than negative. The framing of Section 3—NameRank measures recognition, not merit—is what disciplines the supplemental use: talent-immigration programs exist to identify talent the recognition channel may have missed, so a recognition instrument is by construction not the right primary signal for that target.

7.6 Methodological Robustness

Three methodological choices were tested:

LLM judge necessary, not optional. Pearson(judge, embedding) = 0.15. Embedding alone systematically credits fluent hallucinations on Chinese-name long-tail researchers. Multiplicative coverage×accuracy with LLM judge is the necessary metric structure.

Multiplicative coverage×accuracy necessary, not optional. Per-model refusal patterns show that low-refusal models often hallucinate confidently. A coverage-only or refusal-only metric would over-credit these models. The multiplicative form is what makes confident bluffing strictly worse than refusing.

Open-ended probe necessary, not closed-factoid. Closed-factoid probes have the knows-X-not-Y failure mode. Open-ended probes capture the holistic question "does the model have a representation of this entity?" which is what NameRank is meant to measure.

7.7 Limitations and Open Questions

A compact summary of each caveat, its direction (whether it inflates or deflates the headline findings), and the mitigation available for a future run is given in Appendix I.

1. **Single-probe-per-entity, no test-retest within a model.** We average across 37 models, which provides cross-model stability, but a single probe per (entity, model) pair leaves within-model prompt-sensitivity uncharacterized. A future re-probe with paraphrased templates would quantify this.

2. **Judge family-bias measured at $n = 200$.** The Gemini-judges-Gemini gap is $+0.006 \pm 0.034$. At our 211,603-record scale, even a 0.5pp bias would matter for cohort-level claims, although we are well within calibration uncertainty for the gross findings.

3. **Cohort labels are not fully orthogonal.** Some long-tail-OpenAlex-researcher entities may overlap by name with CS faculty; some IMO golds also appear as CS faculty. We have not de-duplicated across cohorts because the cohort labels serve as descriptive partitions, not exclusive classes.

4. **Chinese-prompt context fields remained English.** The Chinese sub-run translated the probe template but left the disambiguating context in English. A pure-Chinese-prompt re-run would likely amplify the cross-language effects we measure.

5. **Single judge.** We use Gemini 3 Flash Preview as the only judge. A multi-judge consensus would tighten the per-record score and characterize judge disagreement directly. We chose single-judge for cost; the family-bias verification was conducted in a separate check.

6. **Cross-section, not longitudinal.** The data is a 2026-05 snapshot of frontier-model recognition. NameRank evolves with each new model release. A longitudinal re-run at each major release cycle would track propagation of specific names into corpora over time—a direct measurement of how research-discourse-content reaches the LLM training pipelines.

7.8 Predictions for Future Re-Runs

Because NameRank is a snapshot of the current frontier-model fleet's corpus state, a re-run at each major release cycle will produce a longitudinal trajectory. We register three falsifiable predictions for the next-generation NameRank, evaluable when the same probe set is re-run against the model fleet released in 2027:

1. **The credential gap will widen, not narrow.** As frontier models train on more recent corpora, named-artifact-attached careers will accumulate more text and credential-only cohorts will not. *Prediction:* the IMO-gold-vs-OpenAlex-baseline gap grows from -0.102 today to ≤ -0.13 within 12 months.
2. **The artifact $>$ creator inversion will deepen.** New 2026 OSS projects with named-product branding will reproduce the pattern. *Prediction:* the median artifact-minus-creator delta on independent-creator pairs grows from $+0.09$ today (across our 11 pairs) to $\geq +0.14$ within 18 months.
3. **The corpus-density gradient may attenuate at the top.** As Chinese-language training data enters frontier models in greater volume (DeepSeek, Qwen, Kimi, GLM, MiniMax), Chinese-corpus-attached entities will gain modestly. *Prediction:* the Stanford-vs-Tsinghua mean gap narrows by at least 25% but does not close within 24 months.

These predictions are recorded here so that subsequent NameRank revisions can confirm or refute them.

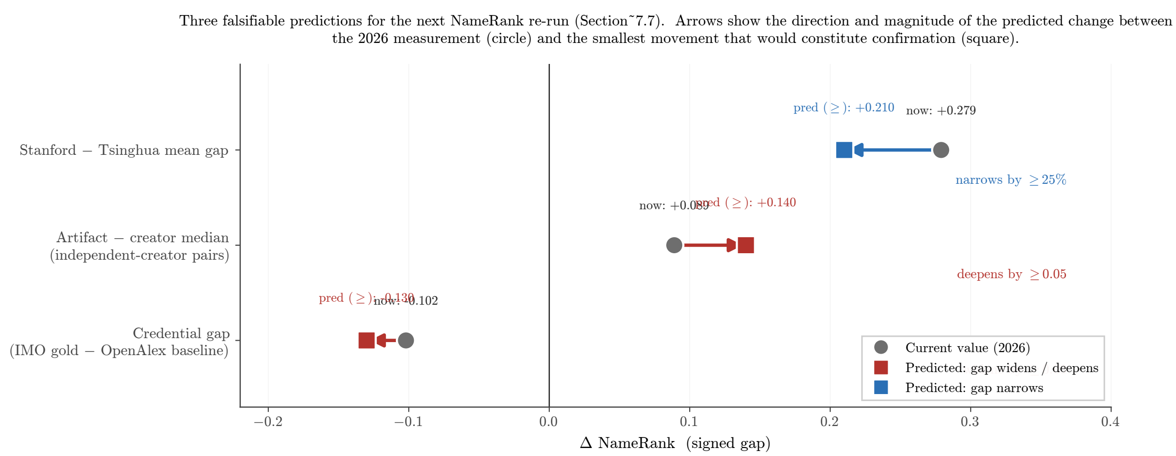

Figure 14: Three falsifiable predictions for the next NameRank re-run. Each row plots a signed gap measured today (circle) and the smallest movement that would constitute confirmation (square). Red arrows mark gaps that widen / deepen under the prediction (credential gap, artifact $>$ creator inversion); blue marks the gap that narrows (corpus-density gradient as Chinese-language training data enters frontier corpora). A re-run on the model fleet released in 2027 will confirm or refute each.

8 Conclusion

We have introduced NameRank, a continuous cross-model recognition metric, and used it to measure 5,719 entities across 54 cohorts via 211,603 probe records plus an 8,880-record Chinese-prompt sub-run. Three findings deserve emphasis. The *credential treadmill thesis* (Section 6.2) shows that once-prestigious intellectual credentials—IMO gold, Rhodes, MSRA Fellowship—no longer translate into name-propagation through the LLM-mediated information channel; they sit at or below the working-researcher baseline. The *artifact > creator inversion* (Section 6.3) shows that for independent creators of named tools, the tool exceeds the creator in NameRank in 8 of 11 verified pairs. The *h-index dominates raw citations* finding (Section 6.4) shows that the structure of citation production matters; its volume does not.

Each of these findings subverts a baseline expectation. The credentialed-meritocracy reader expects IMO gold to remain a strong signal of name-recognition; it does not. The creator-centric reader expects the maker to outrank the made; for independent creators of single named artifacts, the inverse holds. The bibliometrics reader expects citation count to be the right citation feature; it is not—h-index is. Each finding is mechanistically grounded: the credential treadmill traces to lack of post-credential indexable production; the artifact > creator inversion to attribution-direction asymmetry; the h-index dominance to named-attribution-per-paper density.

The broader conceptual claim is that the channel through which human curiosity reaches information about people and artifacts has been re-mediated by a small number of frontier language models, and the propagation of specific names into the corpora that train those models is itself a measurable quantity. Li [2026] found this residual at 65% of recognition variance and characterized its components; NameRank operationalizes the measurement. Three concrete uses for the instrument: (i) as an *impact metric* for cross-domain individuals (a Walmart CTO at 0.27 is now directly comparable to a Stanford CS professor at 0.54); (ii) as a *longitudinal tracker* of how research-discourse content reaches LLM training pipelines (re-run at each major release cycle); (iii) as a *normative indicator* for individual career strategy (the practitioner consequence in Section 7: build and name your own tool).

We release the probe set, gold answers, raw responses, and analysis code at `https://github.com/19PINE-AI/namerank`. The companion website with interactive cohort-distribution explorer, per-entity NameRank lookup, conditional-pair attribution-flow visualizer, cross-language deltas, and a validation page (panel-size sensitivity, per-model generosity/refusal, conditional NameRank): `https://01.me/research/namerank`.

Acknowledgements

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A Full Cohort Specification

Table 8 lists all 54 cohorts with sample size, mean NameRank, refusal rate, and inclusion criteria.

Table 8: Full cohort table. "Refusal" is the fraction of panel responses that returned "unknown" or refusal phrasing. Cohorts sorted by mean NameRank.

Cohort	n	Mean NR	Refusal	Inclusion criteria
mid_tier_gov_ai_policy	42	0.803	6%	US/UK/EU AI-policy government working-level officials
mid_tier_filmmaker	38	0.786	0%	Independent directors with multiple festival features
programming_language	29	0.779	1%	Languages with > 1M users (Python, Rust, Go, etc.)
mid_tier_artist	45	0.774	0%	Contemporary visual artists with major museum representation
database_or_data_system	30	0.773	0%	Open-source databases (Postgres, ClickHouse, etc.)
award	15	0.772	1%	Major academic / industry awards (Turing, Nobel CS-relevant)
benchmark	47	0.747	4%	ML benchmarks (MMLU, GPQA, HellaSwag, etc.)
ai_hardware	20	0.730	3%	AI accelerator products (H100, TPU v5, etc.)
dataset	45	0.725	3%	ML datasets (COCO, ImageNet, etc.)
mid_tier_musician	64	0.692	1%	Working musicians with mid-tier commercial success
mid_tier_chef	47	0.645	1%	Michelin-starred chefs not at the absolute top
ai_startup_or_company	102	0.638	5%	AI startups Series B+ founded post-2018
mid_tier_comedian	46	0.632	2%	Comedians with Netflix specials but not headliners
research_paper	298	0.625	3%	Highly-cited papers (> 10K citations)
mid_tier_historical	105	0.622	0%	Pre-1980 historical figures, less-canonical
conference	30	0.610	0%	CS / ML conferences (NeurIPS, ICML, etc.)
icpc_world_finals_gold	18	0.610	16%	ICPC World Finals gold-medal team members 2005–2015
putnam_fellow	35	0.608	12%	Putnam Mathematical Competition top-25 2005–2015
mid_tier_medical	44	0.603	3%	Mid-career physicians and medical researchers
mid_tier_product	88	0.603	1%	Consumer products with mid-tier adoption
industry_product	9	0.590	4%	Industry products (Cursor, ChatGPT product, etc.)
named_method	73	0.557	1%	Named ML methods (Adam, BatchNorm, Transformer, etc.)

Cohort	n	Mean NR	Refusal	Inclusion criteria
mid_tier_actor	123	0.536	8%	Working actors with multiple credits
mid_tier_founder	48	0.526	4%	Mid-tier startup founders
mid_tier_oss_maintainer	51	0.525	5%	OSS project maintainers
mid_tier_writer	194	0.519	10%	Working authors with multiple published books
oss_project	184	0.519	3%	OSS projects with > 10K GitHub stars
foundation_model	107	0.518	10%	Released foundation models 2020–2026
mid_tier_journalist	84	0.512	2%	Working journalists at major outlets
mid_tier_religious	30	0.486	1%	Notable religious leaders
mid_tier_athlete	134	0.477	10%	Working professional athletes
long_tail_researcher_openalex	**771**	**0.464**	24%	OpenAlex researchers 500–30,000 citations
reference_pilot	37	0.452	15%	C5/C6 diagnostic entities (hand-curated)
website_or_service	9	0.437	10%	Public-utility websites/services
mid_tier_online_course	40	0.434	0%	Notable online courses (CS50, MIT 6.006, etc.)
mid_tier_podcast	57	0.427	2%	Notable podcasts
cs_faculty	**698**	**0.417**	22%	CS faculty from CS Rankings ≥ 1 paper
mid_tier_vc	31	0.402	12%	Mid-tier VCs at top firms
long_tail_paper	**499**	**0.368**	24%	Long-tail academic papers 50–500 citations
noi_china_gold	29	0.368	37%	National Olympiad in Informatics China gold 2005–2015
mid_tier_book	130	0.363	3%	Notable books (literary + non-fiction)
imo_gold	**197**	**0.362**	28%	IMO gold medalists 2005–2015
mid_tier_politician	103	0.356	1%	Mid-tier elected officials globally
msra_phd_fellowship	237	0.343	29%	MSRA PhD Fellowship 2005–2024
ioi_gold	68	0.341	24%	IOI gold medalists 2005–2015
rhodes_scholar	36	0.315	27%	Rhodes Scholarship recipients 2016–2017
cmo_china_gold	131	0.302	38%	China Math Olympiad gold 2005–2012
mid_tier_architect	36	0.255	18%	Working architects without star status
cpho_china_first_prize	50	0.182	53%	China Physics Olympiad first prize 2009
deepseek_v3_author	69	0.178	42%	DeepSeek-V3 paper author list
long_tail_researcher_ikp	159	0.158	47%	IKP Sec. 5.7 researcher subset
mid_tier_activist	2	0.111	15%	Working-level activists ($n = 2$, retained as placeholder; excluded from cohort-level claims)
mid_tier_yc_company	25	0.101	57%	YC company sample from older batches (W19–S22)
gpt5_system_card_author	80	0.042	58%	GPT-5 system card author list

B Probe Template Details

B.1 English Probe Template

```
Tell me what you know about {name}, who is {context}.
If you do not recognize this entity, respond with "unknown".
Limit your response to about 150 words.
```

B.2 Chinese Probe Template (romanized; original Chinese released separately)

The actual probe template used in the Chinese sub-run is the Mandarin translation of the English template, with the literal Chinese characters supplied to the model. We render it here in pinyin (with English translation in parentheses) because the default pdfLaTeX build does not include CJK font support; the literal Chinese version is in the released artifact, and the document can be rebuilt with XeLaTeX to display the characters directly.

```
Qing gaosu wo ni liaojie de guanyu {name} de xinxi,
gai shiti shi {context}.
Ruguo ni bu renshi ci shiti, qing huida "weizhi".
Qing jiang huida kongzhi zai dayue 150 zi yi nei.

(Please tell me what you know about {name},
which is {context}. If you do not recognize
this entity, respond with "unknown". Please
keep your response to about 150 characters.)
```

B.3 Judge Prompt

```
ENTITY: {name}

GOLD ANSWER (authoritative reference, ~150 words):
{gold_answer}

MODEL RESPONSE (to be graded):
{response}

Score the response on two axes:

1. COVERAGE (0.0-1.0): How much of the gold answer's substantive
   content does the response include?
   Substantive content = named artifacts, named affiliations,
   named contributions, identifying facts (year, location, role
   title). Vague statements like "works in AI" or "is famous"
   are NOT substantive content.
   - 1.0: covers all major facts;
   - 0.7: most major facts (>=70%);
   - 0.4: some (>=30%);
   - 0.1: very little (<30%);
   - 0.0: unrelated/refused/unknown.
```

```
2. ACCURACY (0.0-1.0): Are the factual claims in the response
   correct, per the gold answer?
   A "factual claim" is a specific named entity, year, role, or
   relationship asserted by the response. Penalize hallucinations
   heavily.
   - 1.0: all claims correct;
   - 0.7: most correct, minor errors;
   - 0.4: mix of correct and incorrect;
   - 0.1: mostly incorrect (hallucinations dominate);
   - 0.0: entirely fabricated/unrelated.

IMPORTANT:
a) Hallucinated bios (high coverage, low accuracy) MUST
   score LOW on accuracy.
b) Refusals ("I don't know", "unknown") score 0.0 on both axes.
c) Don't penalize correct extra facts that aren't in the gold.
d) Don't credit vague statements that could apply to many.
e) If the gold answer names a contribution and the response
   names it too, count as a substantive fact covered.

Output ONLY JSON: {"coverage": <0.0-1.0>, "accuracy": <0.0-1.0>,
                   "rationale": "<one sentence>"}
```

C Per-Model Refusal and Generosity Statistics

Table 9: Per-model statistics across the 5,719-entity full English run. "Gen." is the mean score per record across all entities; "Ref." is the refusal rate.

Model	Gen.	Ref.	Records
gemini-3-flash-think	0.660	4%	5,719
gpt-5.5-think	0.619	10%	5,719
deepseek-v4-pro-think	0.604	14%	5,719
claude-opus-4.6-think	0.603	10%	5,719
ernie-4.5-300b-a47b	0.603	0%	5,719
llama-4-maverick	0.591	0.1%	5,719
grok-4	0.557	20%	5,719
gemini-2.5-pro-think	0.556	2%	5,719
gemini-3.1-pro	0.551	21%	5,719
deepseek-v3.2-think	0.550	11%	5,719
claude-sonnet-4.6-think	0.542	7%	5,719
gpt-5.4	0.533	18%	5,719
llama-3.3-70b	0.530	5%	5,719
deepseek-v4-flash-think	0.518	26%	5,719
glm-5.1-think	0.517	23%	5,719
glm-4.7-think	0.499	20%	5,719
gemma-3-12b	0.490	7%	5,719
qwen3.5-397b-a17b-think	0.486	1%	5,719
gemma-4-31b	0.483	18%	5,719
glm-4-32b	0.470	20%	5,719
kimi-k2	0.448	31%	5,719
gemma-3-4b	0.443	0%	5,719
kimi-k2.6-think	0.432	38%	5,719
mistral-medium-3.1	0.422	2%	5,719
gpt-5.3	0.399	39%	5,719
phi-4	0.393	0%	5,719
mistral-large	0.389	1%	5,719
llama-3.1-8b	0.374	2%	5,719
grok-4.20-think	0.368	46%	5,719
ministral-3b	0.363	0.1%	5,719
minimax-m2.7-think	0.358	45%	5,719
qwen3-235b-a22b-think	0.308	42%	5,719
qwen3-8b-think	0.286	26%	5,719
qwen3-32b-think	0.284	30%	5,719
gpt-oss-20b-think	0.265	48%	5,719
mistral-small-24b	0.225	46%	5,719
llama-3.2-1b	0.201	0.1%	5,719

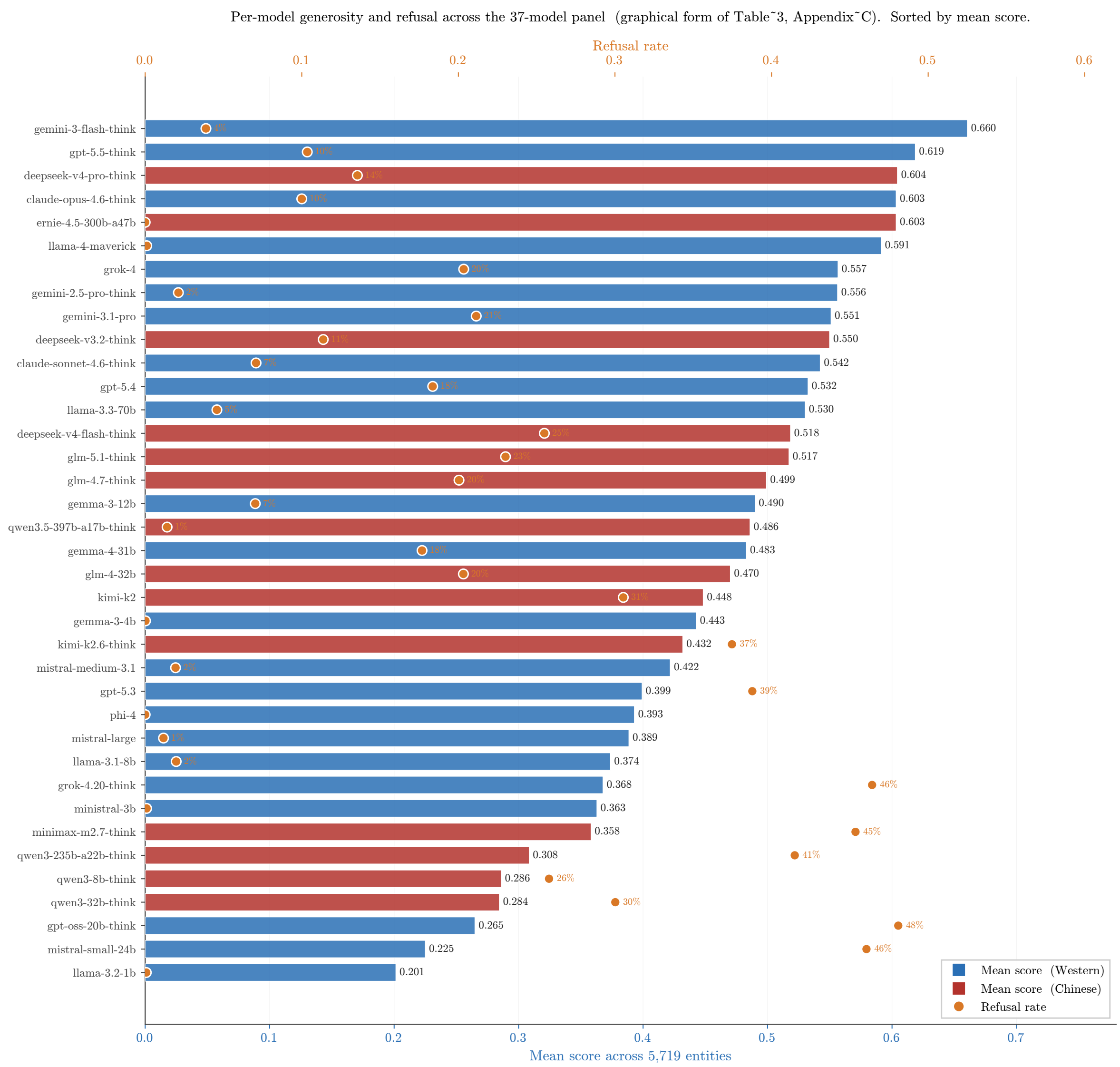

Figure 15: Per-model panel statistics, sorted by mean score (graphical form of Table 9). Bars are mean per-record score across 5,719 entities (lower x-axis, blue/red for Western/Chinese vendor); orange markers on the upper x-axis are the per-model refusal rate, with labels for refusal > 0.001. The two failure regions visible in the main-text Figure 11(b) reappear here at the row level: the strict cluster at the bottom (gpt-oss-20b-think, mistral-small-24b, qwen3-32b/8b-think) refuses heavily; the fluent-hallucinator cluster (ernie, llama-4-maverick, gemma-3-4b, phi-4, ministral-3b, llama-3.2-1b) shows 0% or near-zero refusal, with widely varying mean scores. The panel mean integrates over both styles.

D Conditional-Probe Asymmetry Details

For each of the 11 verified (creator, artifact) pairs, the full per-model breakdown of $C \to A$ and $A \to C$ (fraction of non-refusal model responses that name the counterpart) is released as a supplementary CSV.

Per-model mention table for Jiayi Weng / Tianshou. Of the 28 non-refusal responses to "tell me about Jiayi Weng," 7 name Tianshou: claude-opus-4.6-think, claude-sonnet-4.6-think, gemini-3-flash-think, gemini-3.1-pro, glm-5.1-think, gpt-5.5-think, kimi-k2.6-think. All other models that respond do not name Tianshou. The 7 mentioning models have judge scores ranging 0.70 to 0.80 (mean 0.71); the 21 non-mentioning responding models range 0.00 to 0.50 (mean 0.35). The score lift from artifact-mention is +0.36.

E Cross-Language Per-Entity Deltas

The Chinese-prompt sub-run covers 240 entities. The full per-entity $\Delta(\mathrm{NR_{zh}} - \mathrm{NR_{en}})$ table is released as a supplementary CSV. Selected diagnostic entries:

Entity	$\mathrm{NR_{en}}$	$\mathrm{NR_{zh}}$	Δ	Cohort
Jiayi Weng	0.331	0.327	−0.004	reference_pilot
Tianshou	0.420	0.399	−0.021	reference_pilot
tuixue.online	0.250	0.262	+0.012	reference_pilot
Andrej Karpathy	0.614	0.549	−0.066	reference_pilot
Tri Dao	0.532	0.488	−0.044	reference_pilot
FlashAttention	0.607	0.470	−0.137	reference_pilot
Bojie Li	0.090	0.105	+0.014	reference_pilot
Wang Haoyu (Wang Hao-yu)	0.260	0.430	+0.170	cmo_china_gold
Yao Xuanrong (Yao Xuan-rong)	0.250	0.420	+0.160	noi_china_gold
Wang Zhongjian (Wang Zhong-jian)	0.260	0.420	+0.160	cmo_china_gold
Zijia Zhu	0.100	0.290	+0.190	deepseek_v3_author
Peng Zhang	0.160	0.330	+0.170	deepseek_v3_author

The pattern is consistent: Chinese-character names lift on Chinese prompts; English-coded artifacts lose on Chinese prompts; flat-cross-language entities have name attribution roughly equal in both training corpora.

F External Validity Regressions

Full regression results for NameRank on external impact metrics, restricted to the `long_tail_researcher_openalex` cohort ($n = 771$, citations 500–30,000, h-index from OpenAlex).

Predictor set	R^2	Out-of-sample R^2	Pearson r (held-out)
log(citations)	0.137	0.130	0.379
log(h-index)	0.398	0.428	0.671
log(citations) + log(h-index)	0.398	0.498	0.719

Train/test split is 70/30 random with fixed seed. The marginal R^2 of log(citations) given log(h-index) is 0.000; the coefficient on log(citations) in joint regression is −0.002, indistinguishable from zero.

Decile breakdown of NameRank by h-index (Table 10):

Table 10: Mean NameRank by h-index decile within long_tail_researcher_openalex.

Decile	n	Mean h	Mean NameRank
D1	77	2.9	0.219
D2	77	8.1	0.329
D3	77	12.4	0.392
D4	77	16.7	0.448
D5	77	20.9	0.464
D6	77	25.8	0.524
D7	77	30.6	0.539
D8	77	36.9	0.574
D9	77	45.4	0.546
D10	77	60.4	0.601

The relationship is monotonic and saturates above $h \geq 40$. The bottom decile ($h < 5$) shows NameRank floored at 0.22—the working-researcher floor, not zero.

G Variance Decomposition: Per-Cohort Within-Entity Disagreement

The within-entity model-disagreement (standard deviation across the 37-model panel for each entity, averaged within cohort) ranges over an order of magnitude:

Cohort	Mean within-entity σ
noi_china_gold	0.404
icpc_world_finals_gold	0.381
cmo_china_gold	0.359
long_tail_researcher_openalex	0.356
imo_gold	0.333
putnam_fellow	0.331
...	...
mid_tier_book	0.157
mid_tier_podcast	0.152
named_method	0.144
conference	0.131
gpt5_system_card_author	0.126
mid_tier_online_course	0.113

Interpretation. High within-entity σ cohorts (NOI/ICPC/CMO/IMO/long-tail-OpenAlex) exhibit genuine corpus-pocket-specific recognition: different models trained on different fractions of available corpus see different fractions of the entity's coverage. Low within-entity σ cohorts (online courses, conferences, named methods, books) have universal corpus presence: all models agree because the entity name appears in nearly every training-data source.

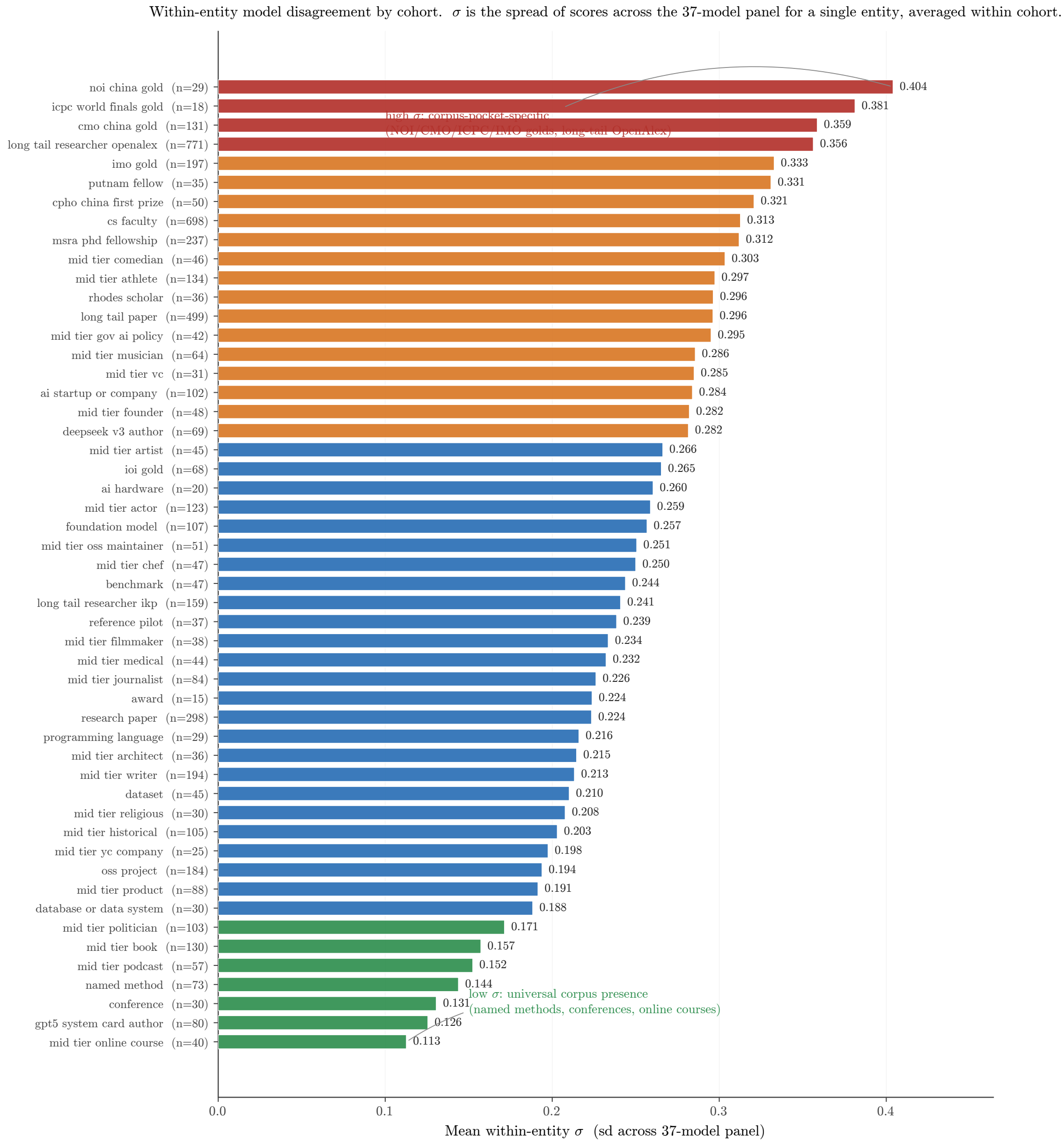

Figure 16: Full within-entity σ ladder across all cohorts with $n \geq 10$. Color bands group cohorts by the four σ regimes (universal $\rightarrow$ corpus-pocket-specific). The high-σ end is dominated by competition-credential and long-tail academic cohorts whose recognition is genuinely corpus-fragment-specific; the low-σ end is dominated by entities whose names appear in nearly every training source (online courses, conferences, named methods, books).

The variance pattern is itself a probe of corpus-pocket distribution: cohorts with high σ are the ones where “which model do you ask?” changes the answer.

H Worked Examples: What a Score of 0, 0.5, and 1 Looks Like

To make the multiplicative coverage×accuracy score concrete, we show three (entity, model, response) triples representing the three regimes the judge identifies.

H.1 Score ≈ 1.0: Andrej Karpathy, asked of gpt-5.5-think

Gold answer (abbreviated): "Andrej Karpathy is a computer scientist and AI educator. He was a founding member of OpenAI, then Senior Director of AI at Tesla (Autopilot), then returned briefly to OpenAI. In 2024 he founded Eureka Labs, an AI education company. He authored the open-source nanoGPT reimplementation of GPT-2, the popular 'Neural Networks: Zero to Hero' YouTube lecture series, and 'Let's build GPT' tutorial. PhD from Stanford under Fei-Fei Li."

Sample response (lightly edited): "Andrej Karpathy is an AI researcher and educator best known for founding Eureka Labs (2024) after stints at OpenAI and Tesla, where he led the Autopilot computer-vision team. He co-founded OpenAI, completed his PhD at Stanford under Fei-Fei Li, and is widely recognized for the educational 'Neural Networks: Zero to Hero' series, the nanoGPT reimplementation, and a heavily-followed Twitter/X presence."

Judge verdict: coverage 1.0, accuracy 1.0, score = 1.0. Six gold-fact items appear (OpenAI, Tesla Autopilot, Eureka Labs 2024, nanoGPT, Zero-to-Hero, Stanford/Fei-Fei Li), all correct.

H.2 Score ≈ 0.5: Aman Sanger, asked of llama-3.3-70b

Gold answer (abbreviated): "Aman Sanger is a co-founder of Anysphere, the company behind Cursor, an AI-powered code editor that became one of the fastest-growing developer tools of 2023-2024. Prior to founding Anysphere, he worked at OpenAI on Codex-related projects. He attended MIT."

Sample response: "Aman Sanger is a software engineer associated with the AI coding tools space. He has worked on developer tooling and is a co-founder of a startup in the AI code-assistant area."

Judge verdict: coverage 0.4 (the company is hinted at but not named), accuracy 1.0 (no false claims), score = 0.4. The response is correct but generic.

H.3 Score ≈ 0 (fluent hallucination): Hao Tang, asked of claude-opus-think

Gold answer (abbreviated): "Hao Tang is a researcher in distributed systems and machine-learning systems. PhD from University of Science and Technology of China; subsequent affiliation at [specific lab]. Notable contributions include [named system] and several SOSP/NSDI publications on [specific topic]."

Sample response: "Hao Tang is a researcher in computer vision and natural language processing. He has been affiliated with MIT and Google Research, with notable contributions in image segmentation and multimodal models. Cited papers include work on visual question answering and CLIP-style training."

Judge verdict: coverage 0.7 (uses the right kind of vocabulary), accuracy 0.0 (every specific claim is fabricated: wrong subfield, wrong affiliation, wrong contributions), score = 0. The embedding similarity with the gold is 0.97 because the response is fluent and on-topic-adjacent.

This is the case that motivates the multiplicative coverage$\times$accuracy structure: coverage alone would credit the response at 0.7; embedding alone would credit at 0.97; only the multiplicative score correctly assigns ≈ 0.

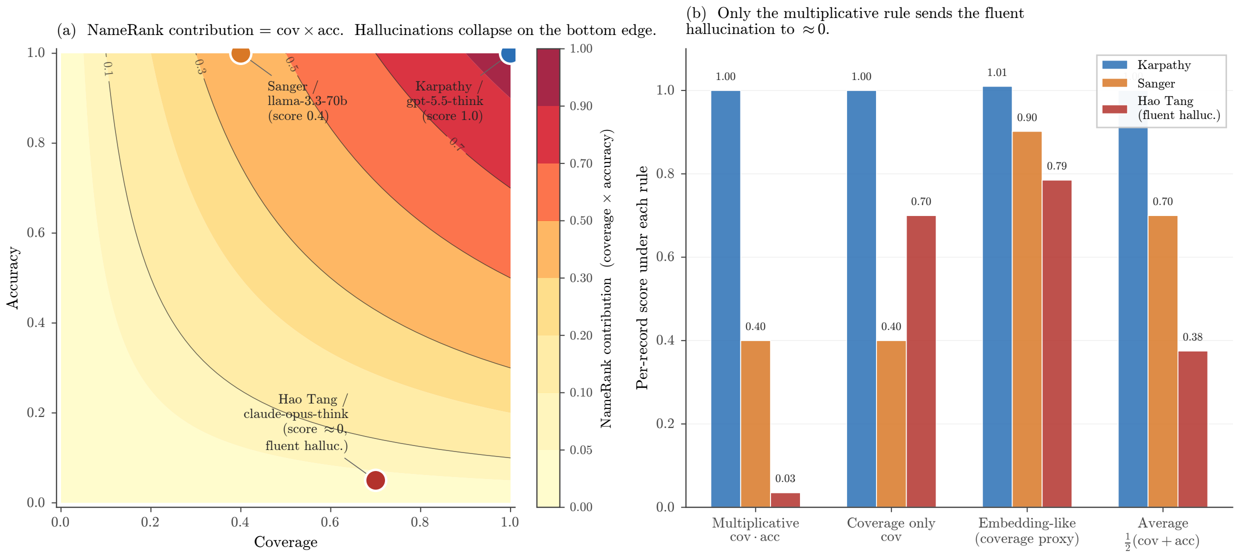

Figure 17: Why the score is multiplicative. **(a)** The (coverage, accuracy) plane with NameRank = $\text{cov} \cdot \text{acc}$ shown as filled contours. The three appendix worked examples sit at the natural exemplar points: a fully-correct Karpathy bio at $(1.0, 1.0)$, a generic-but-correct Sanger summary at $(0.4, 1.0)$, and the Hao Tang fluent hallucination at $(0.7, 0.05)$. **(b)** Per-record score under four alternative aggregation rules. Only the multiplicative rule collapses the fluent hallucination to ≈ 0; coverage-only, embedding-style coverage proxies, and the additive average all credit it at ≥ 0.38.

I Limitations Summary Table

Table 11: Compact summary of methodological caveats, their direction (could overstate or understate findings), and mitigations available for future runs.

Caveat	Direction	Mitigation
Single-probe-per-entity, no within-model test-retest	Inflates within-entity variance	Re-probe with paraphrased templates ($\geq$ 3 paraphrases per entity)
Judge family-bias measured at $n = 200$	Could understate or overstate by $\sim \pm 0.03$ at cohort level	Multi-judge consensus across Gemini + GPT-5 + Claude judges on a $n = 1000$ overlap
Cohort labels not fully orthogonal	Overcounts entities in cross-cohort comparisons	De-duplicate by name within long-tail academic categories
Chinese-prompt context fields remained English	Understates cross-language effect	Pure-Chinese prompts for the full 240-entity sub-run
Single judge (Gemini 3 Flash Preview)	Could carry family-specific scoring style	Replicate with non-Gemini judge on a $n = 500$ overlap
Cross-section, not longitudinal	Cannot infer name-propagation rates	Re-run at each major model-release cycle; release as NameRank v2, v3, ...
Researcher mobility confound in country gradient	Inflates US averages, deflates China averages	Re-classify by country-of-PhD or country-of-doctorate-training as alternative slicing
Tsinghua/Peking small samples ($n = 4$)	Institution-level reads noisy at the bottom	Expand sample by drawing from CCF rank-A faculty rosters
Tianshou case study sample of 1 for per-response mediation analysis	Cannot generalize the +0.36 score lift cleanly	Extend per-response mediation analysis to all 11 verified (creator, artifact) pairs